Certified Joint Pricing Errors in Rough Heston

Residual Propagation and Shared Fourier Errors

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Abstract

How much numerical uncertainty remains in a portfolio of options when every strike is priced from the same approximate transform? In the rough Heston model, separate price intervals lose the common complex error at each Fourier frequency, and keeping it can change a tolerance decision. For a matched quarter-year call spread, the complete joint upper bound is 0.233318843 index points, compared with 0.252393939 from signed marginal aggregation, against a 0.25-point budget. We construct the joint error set by first deriving a residual bound for the fractional Riccati equation and then applying the portfolio pricing functional to the shared transform errors. The propagation kernel is determined by the forward variance curve and the positive fractional resolvent; it retains the full history and the initial curve term. The resulting support formula gives a precise equality condition for the loss incurred by separate price bounds. It also encloses an already computed pricing output, including its displacement from a certified reference, quadrature, the infinite tail and arithmetic. Further experiments separate the effects of history weighting, reference refinement and output replacement. The method certifies numerical error relative to the specified model, which is a separate question from market mispricing.

Keywords: rough Heston; fractional Riccati equation; a posteriori error; portfolio valuation; shared Fourier errors.

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1. Introduction

A portfolio is often valued by adding prices that were computed separately. Its numerical uncertainty need not add in the same way. Options with a common maturity and model parameter use the same characteristic function, so an error in that function enters every strike through a known Fourier coefficient. For a call spread, these coefficients can nearly cancel. A sum of separate absolute error bounds discards this information before the portfolio is formed.

Our question is the following. Given a stored vector of rough Heston option prices, can its error in a specified portfolio direction be certified below a monetary tolerance? The output is held fixed, and the task is to bound its difference from the exact model value; agreement with another numerical solver is not taken as evidence of accuracy. This matters when a valuation has to be checked as delivered, or when several portfolio directions have to be assessed from one common pricing calculation. For the quarter-year call spread studied below, the answer turns on the aggregation alone: with the same node radii, the joint bound certifies a quarter-point budget and signed marginal aggregation does not.

Let c^* be the vector of exact, normalized model prices, c^{fast} the stored output, and $\bar{c}$ a finite reference sum. For a real portfolio vector w, forward F_T, discount factor D, and tolerance τ, the question is whether

$$DF_T \left| w^\top (c^* - c^{\mathrm{fast}}) \right| \le \tau.$$

The reference is useful only if its own uncertainty is controlled. Subtracting and adding that reference gives the exact decomposition

$$c^* - c^{\mathrm{fast}} = d + \mathrm{Re}(A\delta) + r, \qquad d = \bar{c} - c^{\mathrm{fast}}. \tag{1.1}$$

Here A contains the known pricing coefficients and δ_n is the single transform error at frequency u_n, common to all strikes. The remainder r contains the discrepancies outside this finite transform representation. Two questions follow, in this order: how can the transform errors be bounded, and how should those bounds be combined for the portfolio?

1.1. From the Riccati equation to the portfolio error

The transform is determined by a nonlinear fractional Riccati equation. We first specify its unique exact solution and the pricing functional. The fractional derivative prevents the usual ordinary-differential-equation argument from being restarted at an intermediate time: the accumulated history remains part of the equation. For an absolutely continuous reference trajectory, subtraction nevertheless gives an exact linear error equation with a state-dependent coefficient. Its real part is controlled by the left half-plane geometry of the Riccati flow.

The exponent error has a second useful identity. Integration by parts transfers the fractional derivative from the state error to the forward variance curve. Combining this identity with the positive scalar resolvent produces a history weight for the residual. The weight is fixed by the equation and the payoff transform; it is not chosen to reproduce a numerical price. Theorem 3.1 gives the resulting bound, with nonexpansion as a limiting case and a sharper bound when there is positive dissipation.

Once a complex disk encloses each transform error, (1.1) determines a convex set of possible price errors. Evaluating its support in direction w combines the Fourier coefficients before taking their moduli. Theorem 5.1 gives the exact support of this outer set. The equality case of the complex triangle inequality then identifies precisely when separate price bounds lose nothing and when retaining the common errors gives a strictly smaller width. The condition is structural and does not depend on any numerical example.

This leads to a complete certificate for the stored output. The centre displacement d, finite omitted frequencies, strip quadrature, the true infinite tail and arithmetic all remain in the bound. A smaller transform radius alone need not improve the error of the stored output if the reference centre also changes. Keeping these terms explicit is therefore part of the mathematical construction and not an implementation detail.

1.2. The role of approximation

The error theorem is stated for a general reference trajectory satisfying its hypotheses. For computation we use a fractional integral of a piecewise affine function after removing the first startup powers forced by the Riccati equation. The choice does two things: it represents the singular behaviour at time zero explicitly, and it allows continuous residual bounds by exact fractional integration. Section 4 derives the startup coefficients before introducing their stored numerical counterparts. The certificate concerns the full residual on closed time cells; a small residual at sampled nodes is not enough.

The output checked in the principal example uses the established third-order rational approximation of Gatheral and Radoicic. Its endpoint matching gives a compact production formula, and Section 6 proves structural properties on a specified parameter domain. Those properties do not by themselves establish pricing accuracy. The error enclosure comes from the independent reference and is carried over to the production output by (1.1), so the two constructions play different mathematical roles.

The quarter-year spread in Section 7 uses the same stored output, reference centre, node radii and remaining error terms in the joint and marginal calculations. Only the aggregation changes. The complete bounds fall on opposite sides of a quarter-point budget. Other examples expose the limits of the improvement: reference generation has a cost, returning the reference is an alternative when output replacement is permitted, and a numerical-error certificate does not establish agreement with market quotes. The finite-candidate experiment keeps the forward variance curve fixed and reports quote compatibility separately from objective ranking.

1.3. Relation to the literature

Gatheral and Radoicic [GR2019, GR2023] derive rational approximations by matching the short- and long-time behaviour of the fractional Riccati solution. Jeng and Kilicman [JK2020, JK2021] study global Padé approximations and SPX pricing data. We use their approximation framework. The structural results here concern the raw matching system on an explicitly stated domain; they are separate from the residual certificate.

Abi Jaber and El Euch [AbiJaberElEuch2018] provide the Volterra model and affine transform used in Section 2. The analytical ingredients for error propagation are also established: Caputo convexity [LiLiu2018], positive fractional residual comparison [Kopteva2021], Mittag-Leffler positivity [SimonCM2015], and strip trapezoidal analysis [TrefethenWeideman2014]. We apply these results to the model's exponent functional, retain its initial curve term and physical scaling, and propagate the resulting common transform errors to portfolio directions. Appendix C gives the regularity and endpoint arguments required for this composition.

Bayer and Breneis [BBWeak2023, BBSimulation2023] study Markovian kernel approximations and simulation methods. Boyarchenko, de Innocentis and Levendorskii [BL2025] study reliable rough Heston pricing and calibration; their modified Adams method is one of our numerical controls. Ben Hammouda et al. [BenHammouda2026] allocate time-discretization and Fourier-quadrature errors in single- and multilevel methods. Hager and Kreher [HK2026] study expansions in the Hurst parameter. Our question is the a posteriori enclosure of a specified output, complete remainder included. That is a different matter from a complexity bound, an observed convergence rate, or a comparison between nominal solvers.

Section 2 defines the exact problem. Sections 3 and 4 pass from the continuous residual to certified transform values. Section 5 derives the portfolio error set and its financial consequences. Section 6 records the rational output used in the experiments, and Section 7 presents the matched comparisons. Appendices A–E contain the model justification, detailed proofs and reproducibility record.

2. The exact pricing problem

Our conventions for the fractional integral and Caputo derivative are

$$I^\alpha f(t) = \frac{1}{\Gamma(\alpha)} \int_0^t (t-s)^{\alpha-1} f(s)\, ds, \qquad D_C^\alpha v = I^{1-\alpha} v', \quad v \in AC. \tag{2.1}$$

Under the regularity of the trajectories used below, $I^\alpha f \in AC$ with zero initial value, so $D_C^\alpha I^\alpha f = f$. The Caputo operator and all time scalings keep their fractional definitions. With $a = u - i/2$, the physical Riccati state h satisfies

$$D_{C,t}^\alpha h = -\frac{a^2 + ia}{2} + (i\rho\nu a - \lambda_R)h + \frac{\nu^2}{2} h^2, \qquad h(0) = 0. \tag{2.2}$$

Define

$$x = \nu^{1/\alpha} t, \quad y = x^\alpha = \nu t^\alpha, \quad H(x) = \nu h(t), \quad Z(t) = H(\nu^{1/\alpha} t), \quad \kappa = \lambda_R/\nu, \tag{2.3}$$

$$b = (u^2 + 1/4)/2, \quad s_0 = \kappa - \rho/2, \quad d = -s_0 + i\rho u, \quad F(z) = -b + dz + z^2/2.$$

Then $D_{C,x}^\alpha H = F(H)$ and $D_{C,t}^\alpha Z = \nu F(Z)$. Dividing the normalised state error by ν gives the physical h error, and dividing the physical residual $r_t = D_t^\alpha \widehat{Z} - \nu F(\widehat{Z})$ by ν gives the residual bound for the normalised equation.

For the probability model and the numerical examples, we fix $\kappa = 0$ and the entire forward variance curve

$$\xi_*(t) = \theta + (V_0 - \theta)E_{\alpha_0}(-\lambda_\xi t^{\alpha_0}), \quad (\alpha_0, V_0, \theta, \lambda_\xi) = (.5286, .0262, .0721, .5037). \tag{2.4}$$

The curve parameter λ_ξ and Riccati parameter λ_R are defined separately. Curve (2.4) remains fixed when the candidate α changes. The probability model is

$$\begin{aligned} V_t &= \xi_*(t) + \nu \int_0^t K_\alpha(t-s)\sqrt{V_s}\, dW_s, \\ K_\alpha(t) &= t^{\alpha-1}/\Gamma(\alpha), \\ dS_t &= S_t\sqrt{V_t}\, dB_t, \quad d\langle B, W\rangle_t = \rho\, dt. \end{aligned} \tag{2.5}$$

Equation (2.5) is the forward variance representation with zero Riccati mean reversion. Appendix A verifies the probability model and affine transform associated with this curve using Theorems 2.1 and 2.3 and Example 2.2 of Abi Jaber and El Euch.

Let $M_T = S_T/F_T$ be the normalised positive martingale and write $\phi_T(a) = \mathbb{E}M_T^{ia}$. The exact characteristic exponent is

$$L_T(u) = \int_0^T \xi_*(T-t)F(Z(t,u))\, dt, \quad \phi_T(u - i/2) = e^{L_T(u)}. \tag{2.6}$$

With $m = K/F_T, k = \log m, c = C/(DF_T)$, the model European call price is

$$c = 1 - \frac{\sqrt{m}}{\pi}\int_0^\infty \operatorname{Re}\frac{e^{-iuk}\phi_T(u - i/2)}{u^2 + 1/4}\, du. \tag{2.7}$$

The model specification, decimal inputs, and normalisation jointly define the prices considered below. The deterministic propagation theorem allows the more general coefficient in (2.2); the stochastic valuation and experiments in this paper use the zero-mean-reversion forward variance model (2.5).

2.1. The solution to be approximated

The exact object is the solution of the Volterra equation

$$H(x) = \frac{1}{\Gamma(\alpha)}\int_0^x (x-s)^{\alpha-1}F(H(s))\, ds. \tag{2.8}$$

Proposition 2.1 (the exact Riccati trajectory). *Let $\alpha \in (1/2, 1)$, $|\rho| \le 1$, $s_0 \ge 0$, and $u \in \mathbb{R}$. Equation (2.8) has a unique global solution. It is absolutely continuous on every finite interval, locally continuously differentiable at positive times, and satisfies $\operatorname{Re} H(x) < 0$ for $x > 0$. If $s_0 > 0$,*

$$|H(x)| \le \frac{b}{s_0}\{1 - E_\alpha(-s_0 x^\alpha)\} \le \min\left\{\frac{b}{s_0}, \frac{bx^\alpha}{\Gamma(1+\alpha)}\right\}. \tag{2.9}$$

For $s_0 = 0$, the last time-dependent bound remains valid.

Proof. Local Volterra contraction gives existence and uniqueness. On the boundary of the left half-plane, completing the square in $\operatorname{Re} F$ gives a strictly negative term. The fractional maximum principle therefore prevents exit. Convexity of a regularized modulus and comparison with the scalar linear equation give (2.9), which excludes finite-time blow-up. The continuation retains the preceding history. Appendix C.1 proves these steps and the absolute-continuity assertion. □

The proposition identifies the solution independently of any numerical construction. We use its integral representation and its invariant half-plane, and we assume no closed-form solution of the general nonlinear fractional equation. Throughout, existence of the exact solution is kept apart from the construction of an approximation.

2.2. The financial quantity

The state is an intermediate object. The quantity to be bounded is the error after applying the exponent functional (2.6), the Fourier price (2.7), and finally the portfolio vector. A uniform state estimate can discard the time weights in that sequence. The next section keeps those weights by deriving an exact identity for the exponent error before taking absolute values.

3. Finite-history propagation: nonexpansion and dissipation

Let $\widehat{Z}$ be a zero-initial-value reference trajectory and define its physical residual by $r = D_C^\alpha \widehat{Z} - \nu F(\widehat{Z})$. Subtraction of the exact and reference equations gives, for $e = Z - \widehat{Z}$,

$$D_C^\alpha e = \nu \left[d + \frac{Z + \widehat{Z}}{2} \right] e - r, \qquad e(0) = 0. \tag{3.1}$$

This identity uses the quadratic structure of the Riccati equation. The unknown exact state remains in the coefficient, but Proposition 2.1 controls the real part that matters for stability. If $\operatorname{Re} \widehat{Z} \le \epsilon_R$, the coefficient has real part at most $-\nu(s_0 - \epsilon_R/2)$. A bound on the reference half-plane is therefore enough, and no fitted approximation to the exact coefficient is needed.

The second identity concerns the pricing functional. For $\beta > 0$, write $g_\beta(t) = t^{\beta-1}/\Gamma(\beta)$, with all convolutions starting at zero. Define the reference exponent using its fractional derivative, and write $q_\alpha = (I^{1-\alpha}\xi)'$. Under the regularity assumptions of Theorem 3.1, Fubini's theorem and integration by parts give

$$L_T - \widehat{L}_T = \nu^{-1}(q_\alpha * e)(T), \qquad q_\alpha = \xi(0) g_{1-\alpha} + g_{1-\alpha} * \xi'. \tag{3.2}$$

The initial curve value survives the integration by parts and cannot be removed by differentiating the curve only at positive times. The two identities fix the rest of the construction: compare (3.1) with a positive scalar fractional equation, then apply the nonnegative weight in (3.2). Caputo convexity and positive-resolvent comparison are established tools [LiLiu2018; Kopteva2021; SimonCM2015]; Appendix C supplies the regularity and endpoint details needed here.

Define $k_\lambda(t) = t^{\alpha-1} E_{\alpha,\alpha}(-\lambda t^\alpha)$, with $k_0 = g_\alpha$. All convolutions below start at time zero.

Theorem 3.1 (finite-history pricing envelope). *Let $0 < \alpha < 1$, $\nu > 0$, $T > 0$, and $Z, \widehat{Z} \in AC([0,T];\mathbb{C})$ have zero initial value. Let*

$$D_C^\alpha Z = \nu F(Z), \quad r = D_C^\alpha \widehat{Z} - \nu F(\widehat{Z}), \quad |r| \le R \in L^\infty(0,T), \qquad R \ge 0, \tag{3.3}$$

where $F(z) = -b + dz + z^2/2$, $\Re d = -s_0$, $\Re Z \le 0$, $\Re \widehat{Z} \le \epsilon_R$, and $\sigma = s_0 - \epsilon_R/2 \ge 0$. These equations and inequalities hold almost everywhere; the state bounds hold everywhere by continuity. Let $\xi \in AC[0,T]$, $\xi(0) = V_0 \ge 0$, and assume

$$q_\alpha = (I^{1-\alpha}\xi)' = V_0 g_{1-\alpha} + g_{1-\alpha} * \xi' \ge 0 \quad \text{a.e.} \tag{3.4}$$

Define L_T and $\widehat{L}_T$ from $\nu^{-1}\int_0^T \xi(T-t)D_C^\alpha Z(t)\,dt$ and the same expression with $\widehat{Z}$. With $\lambda = \nu\sigma$,

$$|Z - \widehat{Z}| \le k_\lambda * R, \qquad |L_T - \widehat{L}_T| \le \nu^{-1}(q_\alpha * k_\lambda * R)(T) \le \nu^{-1}(\xi * R)(T). \tag{3.5}$$

In particular, $R/\nu \le \delta_F$ implies

$$|L_T - \widehat{L}_T| \le \delta_F \int_0^T \xi(s)\,ds. \tag{3.6}$$

Proof route. Set $e = Z - \widehat{Z}$; the physical equation's divided-difference coefficient has real part at most $-\nu\sigma$. The AC convexity lemma applied to a smooth modulus and the positive zero-initial inverse gives the state bound, including its continuous endpoint. Absolute Fubini and AC integration by parts then express the exponent difference as $\nu^{-1}(q_\alpha * e)(T)$, retaining the zero state initial values and the curve term $V_0 g_{1-\alpha}$. Positivity of q_α propagates that state bound, and $0 \le q_\alpha * k_\lambda \le \xi$ gives the curve envelope. The argument keeps the entire fractional history and removes ν^{-1} only upon normalizing the physical residual by ν; Appendix C.1–C.2 gives the full proof.

At $\sigma = 0$, set $k_0 = g_\alpha$; the two exponent envelopes coincide. Positive dissipation permits further reduction. The initial term $V_0 g_{1-\alpha}$ and the factor ν^{-1} remain in both bounds. Stochastic admissibility of the experimental curve is verified separately in Appendix A.

For a full-history residual envelope $R \le R_j$ on each closed cell $[a_j, b_j]$, the computable cell-weight form is

$$|L_T - \widehat{L}_T| \le \nu^{-1}\sum_j R_j \int_{a_j}^{b_j} (q_\alpha * k_\lambda)(T-s)\,ds \le \nu^{-1}\sum_j R_j \int_{a_j}^{b_j} \xi(T-s)\,ds. \tag{3.7}$$

Every weight integrates the history from time zero, and an integration bin takes the maximum over every closed residual cell it intersects, boundary cells included. Section 7 compares curve and dissipative-kernel weights on matched inputs. Appendix C.3–C.4 proves decreasing-curve and constant-curve variants and a complementary global-state bound.

3.1. The pricing kernel and its computation

Theorem 3.1 determines the pricing kernel by convolution. Its dependence on the forward curve and dissipation is explicit, and its cumulative weights can be enclosed directly. For $\sigma \ge 0$, set $\lambda = \nu\sigma$ and

$$K_\lambda = q_\alpha * k_\lambda, \qquad \eta_{\mathrm{res}} = \nu^{-1}(K_\lambda * R)(T), \qquad 0 \le K_\lambda = \xi - \lambda\,\xi * k_\lambda \le \xi. \tag{3.8}$$

The assumptions and scaling are those of Theorem 3.1, and the positive fractional inverse follows from [Kopteva2021, SimonCM2015]. We enclose the pricing-kernel weights and use them in the same actual-output certificate.

Zero-damping special case of Theorem 3.1. If $\sigma = 0$,

$$k_0 = g_\alpha, \qquad K_0 = \xi, \qquad |L_T - \widehat{L}_T| \le \nu^{-1}(\xi * R)(T). \tag{3.9}$$

The estimate uses $q_\alpha \ge 0$ and preserves the physical scaling at zero damping. Appendix C gives the AC regularity and comparison proofs; Appendix A treats the stochastic-model hypotheses.

For the fixed curve in (2.4), write $A_0 = \alpha_0$ and λ_ξ for its own curve parameter. On the computational domain $\lambda T^\alpha < 1$, define

$$W_\lambda(t)=\int_0^t K_\lambda(s)\,ds=\sum_{n=0}^{\infty}(-\lambda)^n t^{1+n\alpha}\left[\frac{\theta}{\Gamma(2+n\alpha)}+(V_0-\theta)E_{A_0,2+n\alpha}(-\lambda_\xi t^{A_0})\right]. \tag{3.10}$$

A cell $[a_j,b_j]$ has weight $W_\lambda(T-a_j)-W_\lambda(T-b_j)$; Appendix C gives the absolute series tails and the interval-difference proof. The implementation, with twelve outer terms, sixty-four inner curve terms and 100-bit outward dyadic primitives, is an enclosure of the full Mittag–Leffler kernel functional with explicit truncation and rounding remainders. Ordinary floating-point Mittag–Leffler values do not enter a guarantee.

4. From a reference trajectory to transform disks

4.1. A reference dictated by the startup expansion

The preceding theorem separates the choice of reference from the validity of its certificate. A useful computational reference should make the fractional derivative and its residual accessible on a whole time interval, and it should keep the startup singularity of the exact solution.

Write $G=\nu F(Z)$, so that $Z=I^\alpha G$. For fixed model parameters and Fourier frequency, successive substitution into the Volterra equation gives, as $t\downarrow 0$,

$$G(t)=G_0+A_1t^\alpha+A_2t^{2\alpha}+O(t^{3\alpha}),\qquad G_0=-\nu b, \tag{4.1}$$

where the coefficients are determined recursively by

$$\begin{aligned} B_1&=\frac{G_0}{\Gamma(1+\alpha)}, & A_1&=\nu d B_1,\\ B_2&=\frac{A_1\Gamma(1+\alpha)}{\Gamma(1+2\alpha)}, & A_2&=\nu\left(dB_2+\frac{B_1^2}{2}\right). \end{aligned} \tag{4.2}$$

Indeed $Z=O(t^\alpha)$ locally by Proposition 2.1. Substitution first gives $G=G_0+O(t^\alpha)$, then $Z=B_1t^\alpha+O(t^{2\alpha})$, and successive substitution gives the displayed coefficients. The identity $I^\alpha t^\beta=\Gamma(1+\beta)t^{\alpha+\beta}/\Gamma(1+\alpha+\beta)$ controls each remainder. This is a local finite-order expansion at a fixed frequency; it is neither a uniform all-frequency statement nor a global error estimate for a truncated series.

We therefore represent $\bar G$ by these startup powers and a continuous piecewise affine remainder, and set $\widehat Z=I^\alpha\bar G$. The powers come from the equation, and the piecewise affine part allows explicit integration against the fractional kernel. That part represents the rest of the trajectory and is not a fitted price correction. Its nodal values may come from a numerical solver, but Theorem 3.1 does not assume that they are accurate.

In the implementation, all saved nodes and coefficients, including the startup coefficients, are interpreted as exact dyadic numbers. They need not equal their ideal analytical counterparts. The full residual

$$r_t=\bar G-\nu F(I^\alpha\bar G)$$

includes these coefficient discrepancies. Appendix D.1 bounds it on the startup interval and every subsequent closed cell, and verifies the reference half-plane. The certificate belongs to this continuous trajectory and not merely to the nominal solver's nodal states. Four-subcell and two-subcell banks are distinguished in Section 7.

4.2. Completing the transform and price bound

The derivative-based reference exponent is $\widehat{L}_T = \nu^{-1}\int_0^T \xi(T-t)\bar{G}(t)dt$. Once η_n encloses its exponent error, the transform disk follows by the exponential difference and the true half-shift modulus bound. If the substitution-based exponent is $\widehat{L}_T^F = \int_0^T \xi(T-t)F(\widehat{Z}(t))dt$, then $\widehat{L}_T - \widehat{L}_T^F = \nu^{-1}(\xi * r_t)(T)$. Using $\widehat{L}_T^F$ therefore requires this additional conversion term to be enclosed.

$$|\phi_n - \widehat{\phi}_n| \le \min\{1, |\widehat{\phi}_n|\}(e^{\eta_n} - 1). \tag{4.3}$$

Theorem 4.1. *Suppose $M_T > 0, \mathbb{E}M_T = 1$. For any $0 < a_* < 1/2, h_* > 0$, let $g(z) = e^{-ikz}\phi_T(z - i/2)/(z^2 + 1/4)$. Replacing the integral in (2.7) by the infinite trapezoidal sum incurs a price error of at most*

$$\epsilon_{\text{grid}} = \frac{\sqrt{m}\, e^{a_*|k|}}{(1/2 - a_*)(e^{2\pi a_*/h_*} - 1)}. \tag{4.4}$$

Appendix D.2 contains the strip proof and the exact-solution high-frequency comparison. A finite omitted reference value is charged explicitly, independently of the true infinite tail.

Suppose the node values $\widehat{\phi}_n$ have exact-error bounds ε_n. The finite price sum

$$\hat{c}_{N_u} = 1 - \frac{h_*\sqrt{m}}{\pi}\left(2\,\mathrm{Re}\,\widehat{\phi}_0 + \sum_{n=1}^{N_u}\frac{\mathrm{Re}(e^{-inh_*k}\widehat{\phi}_n)}{(nh_*)^2 + 1/4}\right) \tag{4.5}$$

satisfies

$$|c - \hat{c}_{N_u}| \le \epsilon_{\text{grid}} + \epsilon_{\text{tail}} + \frac{h_*\sqrt{m}}{\pi}\left(2\varepsilon_0 + \sum_{n=1}^{N_u}\frac{\varepsilon_n}{(nh_*)^2 + 1/4}\right) + \epsilon_{\text{arithmetic}}. \tag{4.6}$$

In these equations N_u is the Fourier sum cutoff, whereas N_t denotes time-grid cells. The origin coefficient is the trapezoidal half-weight multiplied by $1/(1/4)$. The exact arithmetic remainder encloses the phase, square root, exponential and summation operations.

5. Portfolio errors from a common transform

5.1. The joint error set and the loss from separate bounds

The transform disks are now justified mathematically. What remains is how to use them for several contracts. Fix one model parameter, maturity, contour, and reference grid. Let $c^* \in \mathbb{R}^p$ be the continuous-model price vector, $\bar{c}$ the exact finite reference sum, and c^{fast} the frozen rational output. For each included frequency let $\bar{\phi}_n$ be the reference transform and put $z_n = \phi(u_n - i/2) - \bar{\phi}_n$. Appendix D.1 supplies bounds $|z_n| \le \epsilon_n$; at a deliberately omitted finite node $\bar{\phi}_n = 0$, the true-transform envelope supplies the radius. The same z_n enters every strike at this parameter and maturity.

For the Lewis rule in Section 4, set $m_i = K_i/F$, $k_i = \log m_i$, and

$$a_{in} = -\frac{h\sqrt{m_i}}{\pi}\frac{e^{-iu_nk_i}}{u_n^2 + 1/4} \quad (n > 0), \qquad a_{i0} = -\frac{2h\sqrt{m_i}}{\pi}. \tag{5.1}$$

The half weight at zero is already included in a_{i0}. The complete error is

$$c^* - \bar{c} = \Re \sum_{n=0}^{N} a_{\cdot n} z_n + R, \qquad R \in \mathcal{R}, \tag{5.2}$$

where $\mathcal{R}$ includes the true infinite-grid tail and analytic-strip discretisation error. When a numerical representative of $\bar{c}$ is used, a finite reference sum enclosed by interval arithmetic contributes its centre uncertainty. Omitted finite nodes remain in (5.2), so we do not also charge them as an infinite tail. This representation uses no stochastic covariance.

Theorem 5.1 (complete shared-node inclusion). *Suppose (5.2) holds, all radii are nonnegative, and $\mathcal{R}$ is a nonempty compact convex outer set. Define*

$$\mathcal{E}_F = \left\{ \Re \sum_{n} a_{\cdot n} z_n : |z_n| \le \epsilon_n \right\} + \mathcal{R}, \qquad d = \bar{c} - c^{\mathrm{fast}}. \tag{5.3}$$

Then $c^* - c^{\mathrm{fast}} \in d + \mathcal{E}_F$, and, for real w,

$$h_{d+\mathcal{E}_F}(w) = w^\top d + \sum_{n} \epsilon_n \left| \sum_{i} w_i a_{in} \right| + h_{\mathcal{R}}(w). \tag{5.4}$$

If the centre shift is supplied as a box $d \in [d^-, d^+]$, replace the first term by the support of that box. All numerical evaluations of (5.4) must be outward enclosures.

Proof. Substituting the actual transform errors in (5.2) gives inclusion. The product of complex disks is compact and convex and its real-linear image has those properties. A disk $|z| \le \epsilon$ has real-linear support $\epsilon|b|$ for the functional $\Re(bz)$: Cauchy–Schwarz gives the upper bound, attained at $z = \epsilon\overline{b}/|b|$ when $b \neq 0$; when $b = 0$, every feasible point attains it. Independent disks and Minkowski addition give (5.4). Translation gives the signed centre term. A box enclosing d provides a further valid Minkowski outer enclosure even if its coordinates are dependent. □

If $\mathcal{R} = \prod_i [-\rho_i, \rho_i]$, the smallest coordinate box of this same set has support

$$h_{\mathrm{rect}(\mathcal{E}_F)}(w) = \sum_{i} |w_i| \left(\sum_{n} \epsilon_n |a_{in}| + \rho_i \right). \tag{5.5}$$

Consequently its excess over (5.4), before translation, is

$$G_F(w) = \sum_{n} \epsilon_n \left(\sum_{i} |w_i a_{in}| - \left| \sum_{i} w_i a_{in} \right| \right) \ge 0. \tag{5.6}$$

The equality case of the complex triangle inequality, applied node by node, shows that this excess is strictly positive exactly when one positive-radius node has nonzero coefficients $w_i a_{in}$ that do not all lie on a common nonnegative complex ray. Translation does not change widths. The criterion compares the constructed set with its own coordinate box, so the strict improvement it detects is relative to that box, not to every independently available signed price enclosure.

Any separately established signed model-price box $\mathcal{I}$ can be intersected with $\bar{c} + \mathcal{E}_F$, and the actual model vector belongs to the intersection. We need not solve the intersection support problem to use it, because the minimum of the two valid directional upper bounds remains valid, and likewise the maximum of the lower bounds. A computation can thus retain the best signed marginal information while using node cancellation where it improves

the complete bound. Nonemptiness of the mathematical intersection comes from inclusion, rather than from empirical agreement between numerical approximations.

Proposition 5.2 (nonzero omitted-node reference at an unchanged fast output). *Fix the actual fast output, including its zero contribution at omitted finite nodes. For the same finite pricing map, select a possibly nonzero reference ψ_n at those nodes, and prove $|\phi_n - \psi_n| \le \rho_n$. Let $\overline{c}' = c_0 + \Re(A\psi)$, $d' = \overline{c}' - c^{\mathrm{fast}}$, and let $\mathcal{R}$ include the complete strip, true infinite tail, and newly enclosed reference arithmetic. Then*

$$c^* - c^{\mathrm{fast}} \in d' + \{\Re(Az) : |z_n| \le \rho_n\} \oplus \mathcal{R}. \tag{5.7}$$

For a symmetric remainder and direction w, the complete absolute budget uses

$$DF\left(|w^\top d'| + \sum_n \rho_n \left|\sum_i w_i a_{in}\right| + h_{\mathcal{R}}(w)\right) \le \tau. \tag{5.8}$$

For a nonsymmetric remainder, both support directions must be bounded. A changed reference centre does not change the fixed fast output, but it does change d' and requires new centre arithmetic.

Proof. Subtract and add the new finite reference in the exact finite pricing map; its transform error is $z = \phi - \psi$. The complete remainder contains the other discrepancies. Directional support of the shared complex disks gives (5.8). ∎

Radii attached to different centres cannot be minimized directly. Take an old zero-centred certificate $|\phi_n| \le \varepsilon_n^0$ and a new certificate centred at ψ_n. At the new centre the old certificate only gives $|\phi_n - \psi_n| \le \varepsilon_n^0 + |\psi_n|$, so a safe radius is $\min\{\rho_n, \varepsilon_n^0 + |\psi_n|\}$, or one can retain the two transform disks' intersection. A new disk is nested in the old disk only if $|\psi_n| + \rho_n \le \varepsilon_n^0$. Otherwise, independently valid complete price intervals may still be intersected, but same-centre monotonicity is not automatic. For example, $\phi = 0$ lies in $D(0, .1)$ and $D(1, 2)$, but not in $D(1, \min(.1, 2))$. Marginal and joint comparisons have to use the same updated centre, node radii and complete remainder, and any additional reference evaluation and certification is part of the reported cost.

5.2. Directional and finite-objective certificates

For $w = e_i - e_j$, the node coefficient is

$$|a_{in} - a_{jn}| = \frac{h}{\pi(u_n^2 + 1/4)}\left|\sqrt{m_i}e^{-iu_n k_i} - \sqrt{m_j}e^{-iu_n k_j}\right| \quad (n > 0). \tag{5.9}$$

Cancellation survives because we combine the coefficients before taking their modulus. With $b_i = \sqrt{m_i}$, an analytic bound is

$$|b_i e^{-iuk_i} - b_j e^{-iuk_j}| \le |b_i - b_j| + \min(b_i, b_j)\min\{2, |u|\,|k_i - k_j|\}. \tag{5.10}$$

To prove it, separate the difference in amplitudes from the phase difference using the smaller amplitude, then apply $|e^{ix} - e^{iy}| \le \min(2, |x - y|)$. At zero, the spread coefficient is $2h|b_i - b_j|/\pi$, so low-frequency error cancels for nearby strikes even when the individual node radii are large. High-frequency and tail contributions stay in the bound.

A simple limiting case shows the mechanism without any numerical fit. Fix the model, maturity, node radius and frequency, and let two positive normalized strikes coalesce. With $\Delta k = \log(m_j/m_i)$, the pricing coefficients satisfy

$$a_{jn} = a_{in}e^{(1/2 - iu_n)\Delta k}, \qquad |a_{in} - a_{jn}| = |a_{in}|\left|1 - e^{(1/2 - iu_n)\Delta k}\right|. \tag{5.11}$$

The identity also holds at the half-weighted zero node. Thus the shared spread contribution $\epsilon_n|a_{in} - a_{jn}|$ tends to zero, whereas the separate contribution $\epsilon_n(|a_{in}| + |a_{jn}|)$ tends to $2\epsilon_n|a_{in}|$. For a positive radius this latter limit is strictly positive. At coincident strikes the two payoff functions agree, so the cancellation is already in the contract, and the joint representation preserves it. The statement is about each fixed finite node; the complete output bound still includes its centre displacement and remainder.

The implementation below uses rational outward bounds for logarithms, trigonometric functions, square roots and the circle constant, and it encloses the exact coefficient difference before multiplying by the saved radius. The signed centre comes from reference-sum and frozen-output intervals. Interval midpoints taken for display never stand in for the outward endpoint arithmetic used in a decision.

If the target quote m^* is enclosed around a stored centre $\bar{m}$ with $m^* - \bar{m} \in \mathcal{M}$, use $r = c^{\mathrm{fast}} - \bar{m}$ and replace the output-error set by $\mathcal{E} - \mathcal{M}$, which brings in quote-conversion arithmetic with the correct sign. Otherwise the following formulas take m to be exact.

Let $J(e) = (r+e)^\top W(r+e)/(2p)$, where $r = c^{\mathrm{fast}} - m$, $W \succeq 0$, and the complete output error belongs to a compact convex set $\mathcal{E}$. For any trial vector e_0, put $g_0 = W(r+e_0)/p$. Convexity gives the certified lower bound

$$\inf_{e\in\mathcal{E}} J(e) \geq J(e_0) - g_0^\top e_0 - h_{\mathcal{E}}(-g_0). \tag{5.12}$$

This follows by taking the infimum of the supporting affine function $J(e_0) + g_0^\top(e - e_0)$. Feasibility of e_0 is unnecessary for validity; it affects sharpness. A numerical primal minimizer becomes a lower-bound certificate only after its support or dual obligation has also been bounded correctly.

If $M^2 \geq \sup_{e\in\mathcal{E}} e^\top We$, expansion yields

$$\sup_{e\in\mathcal{E}} J(e) \leq J(0) + \frac{h_{\mathcal{E}}(Wr)}{p} + \frac{M^2}{2p}. \tag{5.13}$$

A safe choice is any valid coordinate box $|e_i| \leq b_i$: with W represented exactly, $M^2 = \sum_{ij}|W_{ij}|b_ib_j$ is sufficient. Tighter validated norm bounds may replace it. Maximization of this convex quadratic is not certified by a local stationary point. A supporting function, norm bound, interval subdivision, or valid relaxation must establish the upper direction.

For a finite candidate collection, form valid objective intervals by intersecting (5.12)–(5.13), when used, with the original interval-square bounds in Theorem 5.3. A unique candidate is certified only if its upper endpoint is smaller than every competing lower endpoint. If this condition is absent, return the nonempty set of candidates whose lower endpoint is at most the least upper endpoint. It contains every exact minimizer. Different candidates have different transform errors; (5.4) does not identify their error variables.

We join the model-specific continuous residual certificate to the actual multi-strike pricing output through the established support formula, equality case of the triangle inequality and convexity bounds, preserving every required budget and its centre. The numerical comparison in Section 7 is our evidence for the resulting financial improvement.

For Theorem 5.3 we specialize the preceding general objective to $p = 12, W = I_{12}$ and sum exactly twelve normalized call prices, so $J = \frac{1}{24}\sum_{i=1}^{12}(c_i - M_i)^2$. Quote-conversion half-widths concern the input midpoint arithmetic and are distinct from the resulting objective half-width.

Theorem 5.3 (finite-objective intervals and selection stability). *For each $\alpha_j \in \Theta_{\text{finite}}$, suppose all model prices have certified enclosures*

$$c_i(\alpha_j) \in [p_{ij}^-, p_{ij}^+], \quad B_i \in [b_i^-, b_i^+], \quad A_i \in [a_i^-, a_i^+], \quad M_i \in [m_i^-, m_i^+]. \tag{5.14}$$

Assume $b_i^- \le b_i^+ \le a_i^- \le a_i^+$. Define

$$\ell(x,y) = \begin{cases} 0, & x \le 0 \le y, \\ \min(x^2, y^2), & \text{otherwise}, \end{cases} \quad v(x,y) = \max(x^2, y^2).$$

$$L_j = \frac{1}{24} \sum_i \ell(p_{ij}^- - m_i^+, p_{ij}^+ - m_i^-), \qquad U_j = \frac{1}{24} \sum_i v(p_{ij}^- - m_i^+, p_{ij}^+ - m_i^-). \tag{5.15}$$

Then $J(\alpha_j) \in [L_j, U_j]$. Writing $L_* = \min_j L_j, U_* = \min_j U_j$, the exact finite-set optimum satisfies

$$J_* = \min_{\Theta_{\text{finite}}} J \in [L_*, U_*]. \tag{5.16}$$

For $\varepsilon \ge 0$, every exact ε-near-optimal candidate belongs to

$$\{\alpha_j : L_j \le U_* + \varepsilon\}; \tag{5.17}$$

The condition $U_j - L_* \le \varepsilon$ is sufficient for that candidate to be near-optimal for the exact objective. If

$$g := \min_{j \ne j_0} L_j - U_{j_0} > 0, \tag{5.18}$$

then α_{j_0} is the unique minimiser of the model objective on the finite set. If

$$\min_{\alpha_j \ge .6} L_j > U_* + \varepsilon, \tag{5.19}$$

then every exact near-optimal candidate in the finite set satisfies $\alpha_j < .6$.

Quote consistency is checked separately. If any row satisfies $p_{ij}^+ < b_i^-$ or $p_{ij}^- > a_i^+$, the candidate cannot lie in all quote bands simultaneously. Conversely, if every row satisfies $p_{ij}^- \ge b_i^+$ and $p_{ij}^+ \le a_i^-$, then quote consistency is certified. Overlapping intervals without this inner-band condition retain possible consistency; overlap alone does not certify it.

Appendix D.4 contains the interval-square and finite-selection proof. A continuous parameter optimum, market identification and dependence across candidate errors lie outside this statement.

6. A rational output with a separate structural guarantee

The certificate above can be applied to any stored price output once its displacement from the reference is enclosed. To make the distinction concrete, we use the Gatheral–Radoicic third-order rational construction. Matching both endpoints reflects two analytical constraints: the fractional startup expansion and the large-time Riccati behaviour. The approximation is thus selected by known features of the equation, while its pricing accuracy is assessed separately.

The construction matches three startup and three large-time coefficients. Write $\widehat{H}(y) = P(y)/Q(y)$, $Q = 1 + q_1 y + q_2 y^2 + q_3 y^3$, with $y = \nu t^\alpha$. Appendix B.1 defines the raw matching matrix, determinant and numerator polynomials, proves their exact identities and gives the full sign proof. No numerical division precedes the determinant lower bound.

Theorem 6.1 (full-frequency structure). *For*

$$\alpha \in [13/25, 3/5], \quad \rho = -1489/2000, \quad \kappa = 0, \quad u \in \mathbb{R}, \quad \nu > 0, \tag{6.1}$$

the established matching system (B.4) is nonsingular. Its normalised denominator satisfies

$$\operatorname{Re} q_j > 0 \ (j = 1, 2, 3), \qquad \operatorname{Re} Q(y) \geq 1, \quad |Q(y)| \geq 1 \quad (y \geq 0). \tag{6.2}$$

Moreover, for $y > 0$, $\operatorname{Re} \widehat{H}(y) < 0$. The result holds at every positive time and every finite real frequency. Its domain is the stated parameter set; $\alpha > .6$, a continuous ρ domain, and nonzero κ are outside the scope of this theorem.

Theorem 6.2 (certified correlation persistence). *The conclusions of Theorem 6.1 hold for*

$$\alpha \in [13/25, 3/5], \qquad \rho \in [-744501/10^6, -744499/10^6], \qquad \kappa = 0, \tag{6.3}$$

at every real Fourier frequency and every positive time.

The correlation width quantifies local persistence around the public-data parameter. Appendix B.2 gives the complete closed cover and the derivative/direct fallback, with derivative bounds for the compactified quantities. For prices we use the independent reference residual in Section 4.

Table 1. Dependencies of the rational-output and independent-reference proof chains.

Proof chain	Inputs and conclusion	Role in the price certificate
Rational construction	Endpoint matching, raw determinant and numerator signs; well-defined Padé output and left-half-plane structure	Defines and checks the specified production formula on its stated domain
Independent reference	Fixed field, full continuous residual, dissipation, finite-history exponent and complete Fourier error	Certifies the reference transform and model-price inclusion
Output translation	Stored fast values and independently enclosed reference centre	Converts the reference inclusion into error of the actual output

7. Matched decisions and scientific controls

The experiments address three distinct questions: whether preserving shared errors changes a portfolio tolerance decision; how much the complete bound changes under a different history weight or reference; and whether the resulting model prices agree with the original quote bands. A comparison isolates one mechanism only when its remaining inputs are held fixed.

All experiments use model parameter $\nu = 0.2897$, $\rho = -0.7445$, Riccati mean reversion zero, Fourier step $1/8$, a finite reference/error grid through 128 (1025 nodes), 100-bit outward dyadic arithmetic and 64 forward-moment series terms. The stored Padé output uses composite Gauss–Legendre order 8 through 200 (352 nodes) and Jacobi order 256; its Fourier cutoff is distinct from the certificate cutoff. N_t counts source time cells. The number of frequency nodes is written explicitly to avoid confusing it with ν.

Table 2. Fixed configurations: maturity in years; time cells, reference nodes, residual subcells and history bins are counts.

ID	T	alpha	N_t source cells	nonzero ref nodes	zero finite nodes	subcells	history bins
H0	1/2	13/25, 3/5, 9/10	2048 / 2048 / 1024	513	512	4	0
H1	1/2	13/25, 3/5, 9/10	2048 / 2048 / 1024	513	512	4	1
H2	1/2	13/25	2048	513	512	4	128
H3	1/2	13/25	2048	1025	0	4	low128; high1
H4	1/2	13/25	2048	1025	0	4	128
Q0	1/4	13/25, 3/5, 9/10	1429 / 1352 / 549	513	512	4	1
Q1	1/4	13/25	1429	1025	0	4	1
Q2	1/4	13/25	1429	1025	0	4	128
N1	1/2	13/25, 21/40, 53/100, 27/50, 11/20	1024	513	512	2	1
N2	1/2	13/25, 21/40, 53/100, 27/50, 11/20	2048	513	512	2	1
N2L	1/2	13/25	2048	513	512	2	128

H0 uses uniform-state propagation; H1 uses finite-horizon global propagation on the same field. H2 uses the low-frequency 128-bin bank. H3/H4 extend the reference through 128, retaining that low bank and using global/128-bin propagation for the high bank. Q0/Q1/Q2 retain the full history from zero and use exact quarter-terminal interpolation of the half-year field: Q0 has 1430/1353/550 time knots, and Q1/Q2 have 1430 knots. They reuse the larger-horizon continuous banks and recompute quarter moments, output and tails. The alpha=.9 source field has 1025 time knots and 641 stored frequencies through 80; 513 of these are nonzero reference nodes in H0/H1/Q0. The .52/.60 fields have 2049 knots and 1025 stored frequencies through 128.

N1/N2 generate each five-point field and continuous residual bank, using two nonstartup subcells (2047/4095 closed cells). N2L is a separate descriptive 128-bin reuse of N2 at alpha=.52. The nearby-grid objective experiment uses N1/N2. The H low and high banks each have 8189 closed cells (four nonstartup subcells), for 513 and 512 frequencies respectively. Thus the 0.115215934-point H4 certificate and 0.394999331-point N2L certificate use different reference centres, node coverage and residual banks; their difference includes more than 128-bin propagation. BL-core512/1024 are nominal history-stepping controls whose complete guarantees use the identified N1/N2/N2L reference bank.

The H and N field generators use a uniform auxiliary grid y_j from zero to νT^α, followed by $x_j = y_j^{1/\alpha}$ and $t_j = x_j/\nu^{1/\alpha}$. In exact arithmetic this is the graded rule $t_j = T(j/N_t)^{1/\alpha}$. Generation uses binary64 arithmetic and forces the endpoints to zero and one half; certification uses the saved nodes as exact dyadics, rather than identifying them with the analytical rule. For quarter-year restriction it keeps every original knot strictly below $T_q = 1/4$, appends the exact endpoint, and encloses the linear interpolation of the original bracketing field coefficients in rational interval arithmetic. All preceding history is retained.

7.1. Fixed contracts, units and complete ablation

The six-month contract uses twelve strikes $K_i = 3700 + 100i$, $0 \le i \le 11$, forward $F = 4221.86$, discount $D = 1$, and the fixed forward-variance curve (2.4). Prices and task errors are in index points, with position multiplier one. Candidate changes affect α while the curve and other parameters remain fixed. The three-candidate profile $.52, .60, .90$ compares this finite set; each candidate is incompatible with several market bid/ask bands. No exchange-specific currency notional is assigned.

Table 3 compares the complete $K = 4400$ minus $K = 4500$ task bound. Each row retains its stored output, signed centre, included finite frequencies, strip, true tail and arithmetic. Displayed upper bounds are rounded upward. The quarter-year study recomputes the reference integral, true-transform envelopes and tails at the shorter maturity and restricts the full-history certificate to that horizon.

Table 3. H0–H4 and Q0–Q2 complete unit-spread certificates, in index points; outputs and reference changes are identified by configuration.

Maturity and stage	Complete joint bound, points	Matched signed marginal bound, points	Interpretation
H0: Six months: global state propagation	1.397612096	—	One-point budget not met
H1: Six months: finite history, same global residual	0.378598956	—	Principal propagation improvement; unchanged output and centre
H2: Six months: local envelope, same reference nodes	0.367258782	—	About 3% further bound reduction
H3: Six months: all finite high nodes, global envelope	0.132245064	0.158585551	High-frequency certification and centre change are included
H4: Six months: all finite high nodes, local envelope	0.115215934	0.137896018	Both methods certify 0.25 points
Q0: Three months: through 64, global envelope	1.207557898	1.545191542	0.25-point budget not met
Q1: Three months: through 128, global envelope	0.351318692	0.408293698	0.25-point budget not met
Q2: Three months: through 128, local envelope	0.233318843	0.252393939	Only the joint method certifies 0.25 points

Finite-history propagation reduces the H0 complete bound by about 72.91%. This measures a guaranteed upper bound. The six-month high-frequency refinement also changes the centre from approximately -0.166762218 to -0.081698202 points, so its reduction combines centre and uncertainty changes. In Q2, the output, centre, node radii and remainders are matched; retaining the shared errors changes the decision. The time-local quarter-year refinement was specified after the global budget failed and is an exploratory transfer experiment.

7.2. Twenty-eight portfolio directions

Let e_i select strike K_i. We keep the exact holdings below, without rescaling them to make a budget pass. Gross weight means $\sum_i |w_i|$. Every threshold is an absolute error in index points for the specified holding vector.

Table 4. The 28 fixed portfolio directions on the twelve strikes; gross weight is the sum of absolute position weights.

Family	Exact direction	Count	Gross weight
Adjacent spreads	$e_i - e_{i+1},\ 0 \le i \le 10$	11	2
Adjacent butterflies	$e_i - 2e_{i+1} + e_{i+2},\ 0 \le i \le 9$	10	4
Wide spreads	$e_0 - e_3, e_3 - e_6, e_6 - e_9, e_0 - e_{11}$	4	2
Positive baskets	$\frac{1}{12}\sum_{i=0}^{11} e_i,\ \frac{1}{6}\sum_{i=0}^{5} e_i,\ \frac{1}{6}\sum_{i=6}^{11} e_i$	3	1

The pass counts use configuration H1 at each stated alpha, and all methods in each matched comparison use identical upstream radii. At a 0.5-point budget and $\alpha = .52$, finite-history joint bounds certify 28/28 portfolios, compared with 13/28 signed marginal bounds. At a 0.25-point budget, the counts are 10/28 against 1/28 for $\alpha = .60$, and 20/28 against 13/28 for $\alpha = .90$. At the one-point budget for .52, both finite-history methods certify 28/28. That threshold does not distinguish the shared structure. A valid payoff intersection gives no further tightening in this setting. Failure to certify leaves the budget unresolved.

7.3. Prespecified nearby candidates under fixed quotes

The original twelve half-year quotes, forward F=4221.86, discount D=1, and all other model parameters are fixed. The prespecified design uses alpha={0.520,0.525,0.530,0.540,0.550}, a first layer N_t=1024, and an upgrade of all five candidates to N_t=2048 if any adjacent joint comparison remains unresolved. Every point has a newly generated continuous reference field and a complete closed-time residual bank. The complete error account includes Fourier nodes through 64, all omitted finite nodes through 128, true infinite tails, the strip remainder, reference arithmetic and the fixed quote-conversion intervals.

The objective is the original normalized midpoint loss $J = \frac{1}{24}\sum_i (c_i - m_i)^2$, using normalized calls. Index-point squared units would require multiplication by F^2. Here the tiny target interval halfwidth is only outward arithmetic error in converting the fixed bid/ask-price midpoint; it is not the market bid/ask halfwidth. Ranking does not establish a uniform winner for arbitrary quote targets inside those market bands. A Taylor support enclosure preserves common Fourier disks in its linear term and charges the full coordinate-radius quadratic remainder. Its same-radius marginal comparison uses the identical upstream certificate. Errors across different alpha candidates are not assumed jointly correlated.

Table 5. N1 and N2 five-candidate objective enclosures under the fixed quotes; displayed values are normalized J multiplied by 10^8.

alpha	N_t=1024 joint J ×10^8	N_t=2048 joint J ×10^8
0.520	[4.825094, 6.778350]	[5.293449, 6.024095]
0.525	[5.645378, 7.501414]	[6.097920, 6.798485]
0.530	[6.748974, 8.517584]	[7.187004, 7.859529]
0.540	[9.807435, 11.422182]	[10.218707, 10.839486]
0.550	[14.000658, 15.482012]	[14.387007, 14.960680]

Table 6. N1 and N2 strict separations among the ten pairs of five fixed candidates, using matched joint and marginal inputs.

Generated layer	Joint strictly separated pairs	Matched marginal pairs
N_t=1024	7/10	3/10
N_t=2048	10/10	7/10

The unresolved close pairs in the first layer triggered the prespecified upgrade of all five candidates. The final finite grid has alpha=0.520 as a strict minimum. This is a five-point comparison, not a continuous calibration optimum. All five candidates remain incompatible with at least one original bid/ask row; the finite-set result therefore does not identify a market-calibrated model.

The two layers generate every closed-cell residual entry: 15,754,230 entries, 5,130 reference exponents and 10,250 true-transform envelopes in total. Separate readers check complete coverage and maxima, reconstruct the exponents, coefficients and objectives, and reject deliberate corruptions. They use common rigorous primitives and the generator's residual-derivative bounds.

7.4. Certifying stored and replaceable outputs

A stored-output certificate applies to pricing-library audits, historical calculations and production outputs fixed by the task. If replacement is allowed and the reference is available, returning its certified centre gives a direct control. Correcting the fast output by that centre also requires the rounding of the stored correction and final addition. The experiments compare these choices. The reference construction and verification must be included when assessing an online workload.

A separate descriptive control, with its configuration fixed in advance, reuses the complete $N_t = 2048, \alpha = .52$ residual bank with 128 history bins. Every intersected closed source cell contributes to each bin maximum, and all weights are outwardly enclosed. All five methods below use the same model, six-month $4400 - 4500$ spread, 0.25-point tolerance, reference centre, node radii, strip and true-tail budget. This control does not change the prespecified nearby-grid experiment.

Table 7. N2L stored-output controls for the half-year unit 4400–4500 spread; complete bounds and the budget are in index points.

Output	Complete joint, points	Matched marginal, points	Joint 0.25-point decision
Stored Padé output	0.394999331	0.514096070	Unresolved
Direct reference (binary64)	0.228237113	0.347333852	Pass
Fast + stored correction	0.228237113	0.347333852	Pass
BL core, 512 steps	0.229082759	0.348179497	Pass
BL core, 1024 steps	0.228534686	0.347631424	Pass

All five matched marginal bounds remain above 0.25 points. Direct reference and corrected fast values have the same complete bound in this instance, with their stored values checked separately. Exact dyadic arithmetic accounts for rounding in the reference return, stored correction and final addition. For these freely replaceable outputs, the reference is simpler than delivering the unchanged fast result. The joint structure still changes the decision for the reference and BL outputs. The BL 512-to-1024 difference is a diagnostic, not its certified error.

Certificate generation encloses 2,100,735 continuous frequency-by-cell residual entries, 513 reference exponents and 1,025 true-transform node bounds, together with the infinite tail and 12,300 price coefficients. Reading checks every entry and reconstructs the exponents, coefficients and decisions. An additional portfolio needs 1,025 shared-disk support terms and the complete remainder. The 28 portfolios reuse the same bank at one parameter and maturity.

Table 8. Typed deterministic work for the additional output constructions; these counts do not measure total computational cost.

Additional output construction	Deterministic mathematical work
Stored Padé output	352 Fourier values; 256 Jacobi samples per frequency
Direct returned reference	Return the already generated, fully charged centre with binary64 rounding
Corrected fast output	Store one correction and perform one binary64 addition per price
BL core, 512 steps	67,371,264 history scalar-vector products; 2,626,560 transformed Riccati evaluations
BL core, 1024 steps	269,222,400 history scalar-vector products; 5,253,120 transformed Riccati evaluations

These counts measure different operations. Reference generation, outward transcendental evaluations and verification are recorded separately from nominal-output construction. The complete ledger also includes exponent quadrature and the two failed coarse-envelope stages. Actual online cost has not been measured.

7.5. Matched dissipative-kernel evaluation

The six-month configuration was fixed at $\alpha = 13/25$, $\nu = 2897/10000$, $\rho = -1489/2000$, and $\sigma = 1489/4000$ before computing the resolvent weights. The physical damping is therefore $\lambda = \nu\sigma = 4313633/40000000$. All four comparisons reuse the same N2L (N_t=2048) closed-cell residual bank, the same stored binary64 fast prices, the same reference exponents and signed centres, and the same finite omission, analytic strip, true infinite tail and reference arithmetic terms. The financial direction is the unit 4400–4500 call spread; the full finite grid contains 1025 frequencies, including 513 used reference frequencies and 512 retained finite omitted frequencies. The complete twelve-price certificate is reconstructed for every method.

For N2L, the dimensionless damping is $\lambda T^{\alpha} \approx 0.075205$ (about 0.075205153821). Dissipation is weak over this horizon, consistent with the modest kernel gain. The complete budget also contains the fixed centre, finite omitted nodes and other remainders.

Table 9 reports upper endpoints in index points, rounded upward to nine decimals. “Global state” first bounds the state by its global residual and then applies the exponent functional; “global curve” propagates that same residual maximum directly through the curve. The final two rows use the identical 128 binwise envelopes and differ only in the pricing kernel.

Table 9. N2L matched propagation for the half-year unit 4400–4500 spread, in index points; the residual bank and all other inputs are fixed.

Propagation	Frozen fast marginal	Frozen fast joint	Stored reference joint	Used-node joint radius
Global state	3.690877564	1.829428107	1.662665889	1.476801373
Global curve	0.589267065	0.428570987	0.261808769	0.075944253
Local curve	0.514096070	0.394999331	0.228237113	0.042372598
Local dissipative resolvent	0.509733613	0.393082828	0.226320610	0.040456094

The binary64 corrected-fast output has the same certified bounds as the stored reference in this configuration, including conversion and rounding. The exact rational reduction from local curve to local resolvent is positive and approximately 0.001916504 points: about 0.485% of the complete frozen-fast bound and 4.523% of its used-node joint radius. These percentages compare upper bounds. The one-point and half-point frozen-output tasks pass; the quarter-point task remains unresolved. Both local methods certify the stored reference and corrected output at a quarter point, so dissipation does not change that decision.

The fixed terms limit further reduction. Even if all 513 used reference-node contributions vanished, the same symmetric budget would retain the absolute centre, finite omitted-node radius and other remainders. Their exact rational sum is strictly greater than the downward decimal endpoint 0.352626733 points. This lower limit belongs to the stated budget formula. With those terms fixed, refining used nodes alone cannot certify a quarter point; the omitted finite frequencies and reference centre must also be addressed.

The reconstructed global-curve and local-curve endpoints agree exactly with the saved baseline rational endpoints, including all twelve individual radii and frozen-output bounds; the table displays strict nine-decimal upward endpoints.

Verification scope and mathematical work

The separate reader reads all 2,100,735 entries of the complete residual bank, checks closed-cell coverage from zero, and recomputes the maxima for all 128 propagation bins. It independently expands the cumulative resolvent by the direct double Gamma series with fourteen outer terms, while the producer uses twelve outer terms and a strict power-field moment. Both enclose all inner tails with sixty-four retained terms. The reader checks all 513 transform-radius minima in all four comparisons and rebuilds all twelve prices with all 1025 finite frequencies. At $\sigma = 0$, it replays all 129 cumulative endpoints and all 128 cell weights against the curve-kernel conclusion. The reader rejects eight negative controls: negative damping, an incorrect physical ν scale, a missing initial closed cell, a removed finite omitted node, an erased true infinite tail, an altered signed centre, an understated resolvent exponent enclosure, and a zero-damping weight that does not reproduce the curve kernel.

The reader reconstructs the cumulative-weight algebra and certificate assembly. It shares the dyadic arithmetic, Gamma, logarithm and exponential primitives, together with the validated residual-derivative bounds and reference exponents. The execution receipts record these dependencies.

This propagation reuses the residual bank. The weighted sum contains $513 \times 128 = 65,664$ terms and 129 cumulative endpoint enclosures, with twelve outer series terms per producer endpoint and sixty-four inner curve terms. These counts describe that calculation; workloads that generate residuals or reference prices perform additional work.

The kernel evidence component contains the fixed contract, interval outputs and separate reader.

7.6. Matched ledgers and certificate resolution

The quarter-maturity decision is attributable to shared aggregation. Q2 uses identical stored output, reference centre, 1025 node radii and all remainders in both columns. Finite omitted-node support is zero because every finite node has a nonzero reference. Values below are in index points; exact fractions are retained in quarter-exact-ledger.json.

Table 10. Q2 quarter-year unit-spread ledger, in index points; only aggregation of the shared node errors differs between columns.

Component	Joint	Signed marginal
Signed centre	-0.149775146827	-0.149775146827
Absolute centre charge	0.149775146827	0.149775146827
Used-node support	0.045461793432	0.064536889507
Finite omitted-node support	0.000000000000	0.000000000000
Strip remainder	0.000026654230	0.000026654230
True infinite-tail remainder	0.038055248032	0.038055248032
Reference arithmetic	4.148871871727e-16	4.148871871727e-16
Complete upper bound	0.233318843	0.252393939

The joint complete bound is 0.233318843 points versus 0.252393939 points for signed marginal aggregation; only the joint certificate passes the 0.25-point budget. The difference 0.019075095 points is entirely the node-support aggregation difference. Complete-bound displays are rounded upward to nine decimal places; component/centre and gap displays are approximate. Every decision and ledger identity uses exact fractional endpoints. Tiny reference arithmetic is displayed separately, and the signed centre is a locator, not an extra additive charge on top of its absolute value.

Every portfolio curve uses the 28 originally fixed and fully specified directions. Exact endpoints, all threshold breakpoints and pass counts are supplied in portfolio-exact-thresholds.csv and portfolio-budget-counts.json. Certification uses the complete rational endpoint condition $B \leq \tau$, including equality. Between-bank plots show descriptive changes; joint versus signed marginal within one bank uses matched radii. No 28-direction quarterly dataset is inferred from the single quarter spread.

Table 11. Complete-budget pass counts for all 28 fixed half-year directions; budgets are in index points and equality is included.

Bank	alpha	Budget points	Marginal	Joint
H0	13/25	1/4	0/28	0/28
H0	13/25	1/2	0/28	0/28
H0	13/25	1	0/28	3/28
H1	13/25	1/4	0/28	2/28
H1	13/25	1/2	13/28	28/28
H1	13/25	1	28/28	28/28
H1	3/5	1/4	1/28	10/28

Bank	alpha	Budget points	Marginal	Joint
H1	3/5	1/2	22/28	28/28
H1	3/5	1	28/28	28/28
H1	9/10	1/4	13/28	20/28
H1	9/10	1/2	25/28	25/28
H1	9/10	1	27/28	28/28
N1	13/25	1/4	0/28	0/28
N1	13/25	1/2	0/28	4/28
N1	13/25	1	15/28	28/28
N2	13/25	1/4	0/28	0/28
N2	13/25	1/2	6/28	23/28
N2	13/25	1	28/28	28/28
N2L	13/25	1/4	0/28	1/28
N2L	13/25	1/2	11/28	26/28
N2L	13/25	1	28/28	28/28

The two nearby levels retain all ten unordered pairs, including unresolved ones:

Table 12. Complete unresolved-pair lists for N1 and N2 under the fixed quotes, alongside strict separation counts.

N_t	Method	Separated pairs	Complete unresolved list
1024	joint	7/10	(0.520, 0.525), (0.520, 0.530), (0.525, 0.530)
1024	marginal	3/10	(0.520, 0.525), (0.520, 0.530), (0.520, 0.540), (0.525, 0.530), (0.525, 0.540), (0.530, 0.540), (0.540, 0.550)
2048	joint	10/10	none
2048	marginal	7/10	(0.520, 0.525), (0.520, 0.530), (0.525, 0.530)

The closest N2 joint separation, alpha=.520 versus .525, is a positive gap of approximately 7.382643478687e-10 in normalized squared-midpoint loss. Its endpoint identity is $J_{0,B} - J_{0,A} - H_B - H_A - Q_A$ plus any box-intersection endpoint adjustment. The gap decomposition, including used nodes, finite zero nodes, reference strip/tail/arithmetic and midpoint-conversion arithmetic, is recorded exactly in nearby-pair-resolution.json. This is a fixed original midpoint-loss ranking. The midpoint-conversion arithmetic interval does not represent the market bid/ask width; all five candidates remain separately bid/ask-incompatible.

The unchanged stored half-year 4400–4500 output admits the following complete centre accounting. Signed centres are approximate displays; radius and complete-bound columns are decimal upper endpoints, so independently rounded columns need not add exactly:

Table 13. Half-year frozen-output centre accounting for H2–H4, N2 and N2L, in index points; upper-endpoint rounding is applied independently.

Bank	Signed centre points	Complete radius points	Complete bound points
H2	-0.166762218	0.200496564	0.367258782
H3	-0.081698202	0.050546863	0.132245064

Bank	Signed centre points	Complete radius points	Complete bound points
H4	-0.081698202	0.033517732	0.115215934
N2	-0.166762218	0.261808769	0.428570987
N2L	-0.166762218	0.228237113	0.394999331

Expanding H2 to H3 changes the absolute centre charge as well as the certified reference radius; H3 to H4 retains that new centre. N2 to N2L retains its own reference centre and full remainders. H4 to N2L is a descriptive cross-bank identity, not an ablation. These transitions are exact in frozen-output-centre-account.json, which also reports the common strict-reference certificate floor as a fraction of each BL-core bound. A dominant floor limits what this comparison can establish about intrinsic solver accuracy. When replacement is permitted and the strict reference is already available, returning that reference directly is the simpler workload; frozen-output audit and free replacement answer different tasks.

7.7. Scope of the experiments

The modern-method comparison reimplements the modified-Adams Riccati core described by Boyarchenko et al.; it does not reproduce their full SINH deformation or Conformal Bootstrap. Agreement between two discretizations is a numerical diagnostic, not an interval certificate. A comparison of complete certified outputs includes the costs of the independent reference, tails, arithmetic and verification.

The fixed refinement menu terminates reliably when the complete task radius meets the budget. Minimum action count assumes that all alternative radii have been certified and each action has unit cost. Constructing those alternatives is part of certificate generation. Model fit, market uncertainty, transaction costs and continuous calibration require additional analysis.

8. Conclusion and reproducibility

The matched spread shows that retaining shared Fourier errors can change a complete tolerance decision without changing the stored prices. It does not show that every portfolio improves, that the upper bound is the actual error, or that a numerically certified model fits market quotes. The finite-candidate comparison is confined to its declared set, and the rational structural theorem is confined to its stated parameter domain. When output replacement is allowed, a certified reference is itself a valid pricing alternative.

The construction follows the valuation map. The Riccati equation determines the exact state and its stability; the forward variance curve determines the residual's history weight; and the portfolio determines the linear functional applied to the common transform errors. Each stage keeps information that an earlier absolute-value bound would lose. The resulting certificate does not depend on how the reference trajectory was generated, provided that its continuous residual, half-plane and arithmetic obligations are verified.

The online article, scientific source and verification commands accompany the paper. All 27 numerical banks are identified by size and SHA-256 in `SCIENTIFIC-MANIFEST.json`. `verify_source.py` checks the publication identities. `reproduce.py` reads the saved banks and reconstructs downstream prices, objectives and decisions; `--full` additionally recomputes the original structural-sign cover. These modes do not regenerate every continuous residual derivative bound. Appendix E and `COMMANDS.md` distinguish the generators, readers and shared arithmetic dependencies.

Appendix A. Probability model and strip transform

We use Abi Jaber–El Euch, arXiv:1803.00477, first posted on 2018-03-01, with a PDF title-page date of March 2, 2018. The journal article appeared in 2019 in Statistics & Probability Letters 149, 63–72. Locators below refer to the 13-page preprint: Theorem 2.1 and Example 2.2 on p.4, Theorem 2.3 on p.5, and H2 and Table 1 on pp.9–10.

For $K_\alpha = t^{\alpha-1}/\Gamma(\alpha)$,

$$\int_0^h K_\alpha^2 dt = \frac{h^{2\alpha-1}}{(2\alpha-1)\Gamma(\alpha)^2}, \quad \int_0^T (K_\alpha(t+h)-K_\alpha(t))^2 dt \le \frac{h^{2\alpha-1}}{\Gamma(\alpha)^2}\int_0^\infty [(r+1)^{\alpha-1}-r^{\alpha-1}]^2 dr. \tag{A.1}$$

The exponent in the latter integral is $2\alpha-2>-1$ at zero and $2\alpha-4<-1$ at infinity, so the integral is finite. The resolvent of the first kind,

$$\mathcal{L}_\alpha(dt) = t^{-\alpha}dt/\Gamma(1-\alpha), \quad K_\alpha * \mathcal{L}_\alpha = 1 \tag{A.2}$$

is verified by the Beta identity and is nonnegative and nonincreasing. The completely monotone spectral measure

$$\mu_\alpha(dx) = x^{-\alpha}dx/[\Gamma(\alpha)\Gamma(1-\alpha)] \tag{A.3}$$

gives $K_\alpha(t) = \int e^{-xt}\mu_\alpha(dx)$. Under $r = xh$, the two spectral integrals in H2 become constants times $h^{\alpha-1}$ and $h^{\alpha-1/2}$, respectively. The remaining constants $\int(1\wedge r^{-1/2})r^{-\alpha}dr$ and $\int r^{-\alpha-1/2}(1\wedge r)dr$ are finite for $.5<\alpha<1$. Table 1 of the source therefore covers the required shifted-kernel conditions.

Curve (2.4) is nondecreasing, has a nonnegative initial value, and is locally Hölder with exponent $\alpha_0 = .5286$. On $\alpha\in[.52,.9]$, the kernel has $\gamma/2 = \alpha-.5 \le .4 < \alpha_0$. Hence the curve-class assumptions in Example 2.2(i) hold, and Theorem 2.1 gives a nonnegative continuous weak variance solution and moment bounds. The equivalent kernel input is

$$b_\alpha(t) = D_C^\alpha \xi_*(t) = I^{1-\alpha}\xi_*'(t) \ge 0, \quad b_\alpha(t) = \frac{(\theta-V_0)\lambda_\xi}{\Gamma(1+\alpha_0-\alpha)}t^{\alpha_0-\alpha} + O(t^{2\alpha_0-\alpha}) \tag{A.4}$$

Its worst exponent is $-.3714>-.5$, so it belongs to L^2_{loc}; also $\xi_* = V_0 + K_\alpha * b_\alpha$. The nonnegative input measure meets Example 2.2(ii). This verifies the model specification with a varying kernel and fixed forward variance curve.

Theorem 2.3 of the source requires $\operatorname{Re}\psi_1\in[0,1]$, a second-state initial transform with nonpositive real part, and a variance forcing term with nonpositive real part. Taking $\psi_1 = ia = 1/2+iu$ and zero for the other initial and forcing terms gives (2.2), (2.6), and uniqueness in weak law. With $\psi_1 = 1$, the Riccati solution is zero and $\mathbb{E}S_T = S_0$ follows. A positive local martingale with constant expectation is a true martingale. Thus the probability strip in (4.4) follows from the continuous-time model. This argument applies established existence and affine-transform theory.

Appendix B. Exact construction and structural certificates

B.1. Matching algebra and full-frequency proof

Set $S = s_0 - i\rho u = -d$, take the principal square root in $A = \sqrt{S^2+2b}$, and let $R = A - S$. The three short-time and long-time coefficients are respectively

$$b_1 = -\frac{b}{\Gamma(1+\alpha)}, \quad b_2 = \frac{Sb}{\Gamma(1+2\alpha)}, \quad b_3 = \frac{\Gamma(1+2\alpha)}{\Gamma(1+3\alpha)}(db_2 + b_1^2/2), \tag{B.1}$$

$$g_0 = -R, \quad g_1 = \frac{R}{A\Gamma(1-\alpha)}, \quad g_2 = -\frac{R}{A^2\Gamma(1-2\alpha)} + \frac{R^2}{2A^3\Gamma(1-\alpha)^2}. \tag{B.2}$$

For $1/2 < \alpha < 1$, $\Gamma(1-2\alpha)$ is finite and negative. The fixed two-endpoint construction of Gatheral and Radoičić [GR2019] requires

$$\widehat{H}(y) = P(y)/Q(y), \quad Q = 1 + q_1 y + q_2 y^2 + q_3 y^3,$$

$$\widehat{H} = b_1 y + b_2 y^2 + b_3 y^3 + O(y^4), \quad \widehat{H} = g_0 + g_1 y^{-1} + g_2 y^{-2} + O(y^{-3}). \tag{B.3}$$

These six conditions give the linear system

$$\begin{pmatrix} g_0 & g_1 & g_2 \\ b_1 & -g_0 & -g_1 \\ b_2 & b_1 & -g_0 \end{pmatrix} \begin{pmatrix} q_1 \\ q_2 \\ q_3 \end{pmatrix} = \begin{pmatrix} b_1 \\ -b_2 \\ -b_3 \end{pmatrix}, \quad p_1 = b_1, \quad p_2 = b_2 + b_1 q_1, \quad p_3 = g_0 q_3. \tag{B.4}$$

Equations (B.1)–(B.4) are the established construction. We next prove invertibility, a nonvanishing denominator, the trajectory half-plane property, and exact-solution error bounds for an independent reference trajectory.

To avoid division by the matching determinant before proving its nonvanishing, define

$$f = \Gamma(1+\alpha)^{-1}, \quad p = \frac{\sin \pi\alpha}{\pi\alpha}, \quad v = -\cos \pi\alpha, \quad m_\alpha = \frac{\Gamma(1+2\alpha)}{\Gamma(1+\alpha)^2}, \quad \zeta = \frac{\Gamma(1+2\alpha)\Gamma(1+\alpha)}{\Gamma(1+3\alpha)},$$

$$\mathsf{M}_2 = Sb/m_\alpha, \quad \mathsf{c} = \zeta(b^2/2 - S^2 b/m_\alpha), \quad V = pR/A, \quad W = m_\alpha p v R/A^2 + p^2 R^2/(2A^3). \tag{B.5}$$

The reflection formula and recurrence relations give $b_1 = -fb, b_2 = f^2\mathsf{M}_2, b_3 = f^3\mathsf{c}$ and $g_1 = V/f, g_2 = W/f^2$. In the latter two expressions, the factors f, f^2 occur in the denominators. Direct expansion of the determinant and Cramer numerators of (B.4) gives

$$\begin{aligned} \Delta &= b^2 W + 2bRV - \mathsf{M}_2 RW - \mathsf{M}_2 V^2 - R^3, \\ F_1 &= b^2 V + b\mathsf{M}_2 W - bR^2 + \mathsf{M}_2 RV + \mathsf{c}RW + \mathsf{c}V^2, \\ F_2 &= -b^2 R + b\mathsf{M}_2 V + b\mathsf{c}W + \mathsf{M}_2^2 W + \mathsf{M}_2 R^2 + \mathsf{c}RV, \\ F_3 &= -b^3 + 2b\mathsf{M}_2 R - b\mathsf{c}V - \mathsf{M}_2^2 V + \mathsf{c}R^2. \end{aligned} \tag{B.6}$$

Here Δ is exactly the determinant of the original system, and the numerators are $N_j = f^j F_j$. After establishing $\Delta \neq 0$, we may define $q_j = N_j/\Delta$, in which case

$$\Delta P = -fb\Delta y + f^2(\mathsf{M}_2\Delta - bF_1)y^2 - f^3 R F_3 y^3, \quad \Delta Q = \Delta + fF_1 y + f^2 F_2 y^2 + f^3 F_3 y^3. \tag{B.7}$$

In physical time, the denominator coefficients are $\nu^j q_j$; the normalised coefficients q_j retain their original definition.

Proof. First take $u \geq 0$. Set

$$\omega = \sqrt{u^2 + 1/4}, \quad \eta = u/(1+u), \quad j(\eta) = \sqrt{\eta^2 + (1-\eta)^2/4}, \quad s = -\rho/2,$$

$$\bar{S} = s(1 - \eta + 2i\eta)/j(\eta), \quad \bar{A} = \sqrt{1 + \bar{S}^2}, \quad \bar{R} = (\bar{A} + \bar{S})^{-1}. \tag{B.8}$$

Since $j^2 \geq 1/5$, these functions are defined on the closed interval $\eta \in [0,1]$. We have $|\bar{S}| = |\rho|$ and

$$\operatorname{Re} \bar{A}^2 = 1 - \rho^2 + 2(\operatorname{Re} \bar{S})^2 \geq 1 - \rho^2 > 0.$$

The principal square root is continuous and lies in the first quadrant; $\operatorname{Re}\bar{A}, |\bar{A}| > 3/5$. The real part of the Hermitian product of two first-quadrant numbers is nonnegative. Hence

$$|\bar{A}+\bar{S}|^2 \geq |\bar{A}|^2 + |\bar{S}|^2 \geq |\bar{A}^2 - \bar{S}^2| = 1, \quad |\bar{R}| \leq 1, \quad \operatorname{Re}\bar{R} > 0.$$

Equation (B.8) expresses $\bar{A}-\bar{S}$ in reciprocal form, with a denominator determined directly by the current parameters.

Substitute $\bar{b} = 1/2$ and (B.8) into (B.5)–(B.6) to obtain the barred original quantities. Homogeneity gives

$$\Delta = \omega^3 \bar{\Delta}, \qquad F_j = \omega^{3+j}\bar{F}_j. \tag{B.9}$$

Define three real functions

$$\bar{B}_j = \operatorname{Re}(\bar{F}_j \overline{\bar{\Delta}}). \tag{B.10}$$

The rigorous rational covering in Appendix B proves that, throughout the closed rectangle $[13/25, 3/5] \times [0,1]$, $\bar{B}_j > 0$ and

$$|\bar{\Delta}|^2 \geq \frac{8811080120918477887462 8745}{1267650600228229401496703205376} > \frac{1}{14400}. \tag{B.11}$$

This first establishes $\Delta \neq 0$ and then yields $\operatorname{Re} q_j = (f\omega)^j \bar{B}_j/|\bar{\Delta}|^2 > 0$, proving (6.2).

For the trajectory half-plane, the finite convolution in (B.7) gives

$$|\Delta|^2 \operatorname{Re}(P\overline{Q}) = \sum_{n=1}^{6} f^n D_n y^n, \tag{B.12}$$

$$\begin{aligned}
D_1 &= -b|\Delta|^2, \\
D_2 &= \operatorname{Re}(\mathsf{M}_2)|\Delta|^2 - 2b\operatorname{Re}(F_1\overline{\Delta}), \\
D_3 &= \operatorname{Re}\{-RF_3\overline{\Delta} + (\mathsf{M}_2\Delta - bF_1)\overline{F}_1 - b\Delta\overline{F}_2\}, \\
D_4 &= \operatorname{Re}\{-RF_3\overline{F}_1 + (\mathsf{M}_2\Delta - bF_1)\overline{F}_2 - b\Delta\overline{F}_3\}, \\
D_5 &= \operatorname{Re}\{-RF_3\overline{F}_2 + (\mathsf{M}_2\Delta - bF_1)\overline{F}_3\}, \\
D_6 &= -\operatorname{Re} R\,|F_3|^2.
\end{aligned} \tag{B.13}$$

The same rational covering certifies the six compactified quantities $\bar{D}_n < 0$, with scaling $D_n = \omega^{7+n}\bar{D}_n$. Set $z = f\omega y$. The right-hand side of (B.12) is $\omega^7 \sum_{n=1}^{6} \bar{D}_n z^n$ and is therefore strictly negative for every $y > 0$. Having proved that the denominator is nonzero, division by $|Q|^2|\Delta|^2$ gives $\operatorname{Re}\widehat{H} < 0$. Negative frequencies follow by conjugate symmetry of the principal root and matching coefficients. The closed endpoint $\eta = 1$ represents the infinite-frequency limit, so the proof on the closed compactified domain covers every finite frequency. ∎

Positive real parts of the coefficients provide a sufficient condition for a nonvanishing denominator. Appendix B specifies the algebraic identities, elementary-function remainder bounds, and continuous interval covering used in the computer-assisted proof.

A continuous, quantitatively bounded correlation extension

The domain contains the public-data correlation and roughness candidates for the fixed six-condition construction. Its sign cover includes a continuous roughness interval and all frequencies. Strict signs imply persistence in correlation; the following bounds quantify that local persistence.

The certificate domain is $\mathcal{B} = [13/25, 3/5] \times [0, 1]$. Every interval $I = [\ell/2^{100}, r/2^{100}]$ uses integer outward rounding. Rational inputs are rounded down and up; products take extrema over the four endpoint corners and are quantised. Reciprocals first exclude zero. Square roots use integer isqrt and an increment for the upper endpoint. Complex quantities use rectangular real–imaginary interval arithmetic. The analytic lower bounds in (B.8) fix the principal-root and reciprocal branches; no floating-point sign tolerance is used.

The four basic α-dependent functions p, v, m_α, ζ satisfy

$$p' < 0, \quad v' > 0, \quad (\log m_\alpha)' = 2[\psi(1+2\alpha) - \psi(1+\alpha)] > 0,$$
$$(\log \zeta)' = 2[\psi(1+2\alpha) - \psi(1+3\alpha)] + [\psi(1+\alpha) - \psi(1+3\alpha)] < 0. \tag{B.14}$$

We have $\psi'(x) = \sum_{n \geq 0} (x+n)^{-2} > 0$, and elementary trigonometric identities give the derivatives of p, v. Strict endpoint values therefore enclose each continuous α interval, without sampling α at grid points.

The constant π is certified by Machin's identity and one hundred terms of each alternating arctangent series. The logarithm is reduced to $[1, 2]$ and evaluated with one hundred positive atanh-series terms and a geometric tail. The exponential is reduced to $[0, 1]$, evaluated with one hundred Taylor terms and a geometric tail, and restored by repeated rigorous squaring. For positive real Gamma arguments, recurrence first shifts to $z \geq 20$. Retaining $B_2, \ldots, B_{20}$ in log Gamma gives the positive-real remainder bound

$$0 < R_{10}(z) < B_{22}/(22 \cdot 21 z^{21}). \tag{B.15}$$

In Binet's positive integral, the ten-term geometric remainder of arctangent is positive and bounded by the first omitted power. Integration gives (B.15), consistent with the positive-real Stirling remainder in NIST2010. After exponentiation, divide successively by the recurrence factors. Sine and cosine retain thirty-two terms, with absolute Lagrange tails $M^{65}/65!, M^{64}/64!$. Bernoulli numbers, factorials, and interval endpoints are exact rationals.

For each closed rectangle, substitute these enclosures successively into (B.8), (B.5), (B.6), (B.10), and (B.13). Accept only when the lower endpoints of all three $\bar{B}_j$ are strictly positive and the upper endpoints of all six $\bar{D}_n$ are strictly negative. Otherwise bisect one coordinate at its exact rational midpoint. The two closed children have the parent rectangle as their union, and strict enclosures also cover the shared boundary.

The finite covering contains 211241 leaf rectangles of total rational area exactly $2/25$. An independent check reconstructs 422481 binary-tree nodes from the root, with maximum depth 19 and each recorded leaf occurring exactly once. In addition to the area check, this excludes missing subtrees, interior overlap, and endpoint gaps. The theorem uses the order interval covered by this complete closed tree.

The stronger rational margins in the complete certificate imply the following bounds. Each simplification has been independently checked using Fraction.

Table B.1. Strict lower bounds on the full compactified structural domain specified in Appendix B.

Quantity on the full compactified domain	Strict lower bound
$\bar{B}_1, \bar{B}_2, \bar{B}_3$	1/6000, 1/8000, 1/20000000000
$-\bar{D}_1, -\bar{D}_2, -\bar{D}_3$	1/30000, 1/2000000000, 1/6000
$-\bar{D}_4, -\bar{D}_5, -\bar{D}_6$	1/8000, 1/15000000, 1/100000
$\lvert\bar{\Delta}\rvert^2$	1/14400

The proof consists of rigorous interval inclusion and a complete finite covering. The scientific source provides the interval implementation, all leaf rectangles, and the tree-structure check for independent verification.

B.2. Continuous correlation persistence

Proof. Retain the frequency compactification $\eta = u/(1+u)$, $0 \le \eta \le 1$, and the original unnormalized polynomials Δ, F_j, B_j, D_j. Write $S_b = -\rho[(1-\eta) + 2i\eta]/(2h)$, $h^2 = \eta^2 + (1-\eta)^2/4$. Then $|S_b| = |\rho|$, $\Re S_b \ge 0$ for negative ρ, $A^2 = 1 + S_b^2$, and $R = (A + S_b)^{-1} = A - S_b$. Throughout $|\rho| \le 3/4$, the square-root branch remains fixed because $\Re A^2 \ge 1 - \rho^2 > 0$. Consequently,

$$|A|^{-1} \le 8/5, \quad |A'| \le 6/5, \quad |(A^{-1})'| \le 384/125, \quad |R| \le 1, \quad |R'| \le 11/5, \tag{B.16}$$

where primes denote real ρ derivatives. The identity $|R| \le 1$ follows from $|A + S_b|^2 \ge 1$: the terms $\Re A \, \Re S_b$ and $\Im A \, \Im S_b$ are nonnegative, while $|A|^2 = |1 + S_b^2| \ge 1 - |S_b|^2$. Also $\Re R > 0$, since $(\Re A)^2 - (\Re S_b)^2 = (|1 + S_b^2| + 1 - |S_b|^2)/2 > 0$.

For the α-dependent scalars of Theorem 6.1, exact outward endpoint enclosures give

$$0 < p \le 2/3, \quad 0 \le v \le 1/3, \quad 1 \le m \le 3/2, \quad 0 < \zeta \le 2/3. \tag{B.17}$$

Here p decreases and v, m increase on the stated interval, while ζ decreases. The gamma-ratio assertions follow from the increasing digamma function; $m \ge 1$ also follows from log convexity of Γ. None depends on ρ.

Apply product-rule modulus bounds to the original polynomial formulas. For every (α, η), the resulting exact rational constants $L_{B,j}, L_{D,j}$ satisfy

$$|\partial_\rho B_j| \le L_{B,j}, \qquad |\partial_\rho D_j| \le L_{D,j}. \tag{B.18}$$

They are assembled using pairs (M, N), meaning $|f| \le M$, $|f'| \le N$, with addition $(M_1 + M_2, N_1 + N_2)$ and multiplication $(M_1 M_2, N_1 M_2 + M_1 N_2)$. This yields a finite, directly checkable rational derivation rather than a numerical derivative estimate.

Let $m_{B,j}(C), m_{D,j}(C)$ be the exact original sign lower bounds on a continuous (α, η) cell C at $\rho_0 = -1489/2000$. A cell is certified throughout the target correlation interval whenever all bounds

$$m_{B,j}(C) - 10^{-6} L_{B,j} > 0, \qquad m_{D,j}(C) - 10^{-6} L_{D,j} > 0 \tag{B.19}$$

hold. The mean-value theorem proves this sufficient test; it uses a derivative enclosure, not numerical differentiation. It certifies 182002 original leaves and 79758 finer leaves. For the remaining cells, directly evaluate the original complex polynomial formulas with ρ, α and η all interval valued. Exact outward dyadic arithmetic certifies strict $B_j > 0$ and $D_j < 0$ on 19808 additional closed cells.

The original complete midpoint tree and the exact local splice trees verify that these 281568 cells cover the entire (α, η) rectangle without gaps or interior overlap. Their exact area is $2/25$; each cell carries the entire correlation interval, giving exact three-dimensional volume $1/6250000$. Hence the raw signs hold for every point of the stated domain. In particular $\Delta \neq 0$, the normalized denominator coefficients have positive real part, and all real numerator coefficients of the half-plane test are negative. The original algebraic implication gives (6.3), including $\eta = 1$, the infinite-frequency limit. Conjugacy supplies negative frequencies. ∎

This is a narrow local robustness result: the total correlation width is 2×10^{-6}. It does not establish a typical broad calibration box or a new price experiment at an altered correlation. The direct interval supplement is explicitly exploratory: its protocol was frozen after the derivative-only attempt left positive unresolved area. The evidence retains that partial result and the separately valid conservative fallback.

The independent reader checks every saved original sign, the original 422481-node partition tree, the local splice geometry, the independently assembled derivative majorants and all 19808 direct cells through separate Cramer matrices and numerator-polynomial convolution. The dyadic primitives and saved original sign generation remain shared dependencies; every old interval-polynomial evaluation is not regenerated. Exact large fractions stay in the machine ledger. Neither a sampled correlation grid nor an assertion of continuity replaces any cell in this proof.

Appendix C. Complete propagation proofs and analytic variants

C.1. Regularity, convexity and dissipative state comparison

If $H = I^{\alpha} F(H)$ is a bounded local continuous solution, then $H, F(H)$ are α-Hölder continuous. For $\alpha > 1/2$, write $q = F(H)$. Cancellation gives the derivative formula

$$H'(x) = \frac{q(x) x^{\alpha-1}}{\Gamma(\alpha)} + \frac{\alpha - 1}{\Gamma(\alpha)} \int_0^x (x-t)^{\alpha-2} [q(t) - q(x)] \, dt. \tag{C.1}$$

The bound

$$|H'(x)| \leq \frac{\|q\|_{\infty} x^{\alpha-1}}{\Gamma(\alpha)} + \frac{(1-\alpha)[q]_{C^{\alpha}} x^{2\alpha-1}}{\Gamma(\alpha)(2\alpha-1)}. \tag{C.2}$$

is integrable at zero. Positive-time continuity and the limit from truncated intervals establish absolute continuity on the closed interval. The near-endpoint exponent satisfies $2\alpha - 2 > -1$; the other endpoint and the first term are integrable. Thus $H \in AC$ and it is locally C^1 at positive times. Local existence follows from Volterra contraction on a bounded ball. For $v \in AC$ that is locally Lipschitz at positive times, integration by parts gives

$$D_C^{\alpha} v(x) = \frac{1}{\Gamma(1-\alpha)} \left\{ \frac{v(x) - v(0)}{x^{\alpha}} + \alpha \int_0^x \frac{v(x) - v(t)}{(x-t)^{1+\alpha}} \, dt \right\}. \tag{C.3}$$

At a positive maximum over the full history, with zero initial value, this derivative is strictly positive. Consequently, $D_C^{\alpha} v + s v \leq D_C^{\alpha} w + s w$, $v(0) = w(0)$, and $s \geq 0$ imply $v \leq w$. For a convex C^1 function Φ on the real plane, apply the supporting-hyperplane inequality to both terms in (C.3) to obtain

$$D_C^{\alpha} \Phi(v) \leq \nabla \Phi(v) \cdot D_C^{\alpha} v. \tag{C.4}$$

This Caputo history-convexity inequality follows from the supporting-hyperplane inequality. Under the regularity assumptions of Proposition 3.11 of Li and Liu [LiLiu2018], we prove the form needed below.

Lemma C.1. *Suppose $\alpha \in (1/2, 1), |\rho| \le 1, s_0 \ge 0$. The normalised equation has a unique global solution, and $\operatorname{Re} H(x) < 0$ for $x > 0$. If $s_0 > 0$, then*

$$|H(x)| \le \frac{b}{s_0}[1 - E_\alpha(-s_0 x^\alpha)] \le \min\{b/s_0, bx^\alpha/\Gamma(1+\alpha)\}. \tag{C.5}$$

When $s_0 = 0$, the latter time-dependent bound remains valid.

Proof. Write $H = X + iY$. Completing the square gives

$$\operatorname{Re} F(H) = -\frac{1}{8} - \frac{1-\rho^2}{2}u^2 - \frac{1}{2}(Y + \rho u)^2 - s_0 X + \frac{1}{2}X^2. \tag{C.6}$$

If X first reaches a small positive level $\varepsilon < 1/2$, the derivative in (C.3) is positive, whereas (C.6) is negative, a contradiction. Thus $X \le 0$. Reaching zero at a positive time gives the same contradiction, proving strict negativity.

Let $\psi_\epsilon(z) = \sqrt{|z|^2 + \epsilon^2} - \epsilon$. Equation (C.4), together with

$$\operatorname{Re}(\overline{H} F(H)) = -bX - s_0|H|^2 + \tfrac{1}{2}X|H|^2, \quad \frac{|H|^2}{\sqrt{|H|^2 + \epsilon^2}} \ge \psi_\epsilon(H),$$

gives $D_C^\alpha \psi_\epsilon(H) + s_0 \psi_\epsilon(H) \le b$. Compare with the zero-initial-value linear scalar equation and let $\epsilon \downarrow 0$ to obtain (C.5). The scalar solution is verified directly by the Mittag–Leffler series. If finite-time blow-up occurred, (C.5) would bound the trajectory, $F(H)$, and a uniform Hölder constant. The history integral would have a finite limit at that endpoint, and local contraction with the previously accumulated history as forcing would extend the solution, a contradiction. The Caputo history is not restarted. Successive Volterra contractions give uniqueness. ∎

Theorem C.2 (dissipative residual bound). *Suppose $s_0 > 0$, $\widehat{H}(0) = 0$, and $\widehat{H} \in AC$, with local Lipschitz regularity at positive times. Assume, for every $x \in (0, X]$, that*

$$\operatorname{Re} \widehat{H} \le \epsilon_R < 2s_0, \qquad |D_C^\alpha \widehat{H} - F(\widehat{H})| \le \delta,$$

Set $\sigma = s_0 - \epsilon_R/2 > 0$. Then

$$|H - \widehat{H}| \le \frac{\delta}{\sigma}[1 - E_\alpha(-\sigma x^\alpha)] \le \delta/\sigma. \tag{C.7}$$

In particular, for a trajectory in the left half-plane, one may take $\epsilon_R = 0, \sigma = s_0$. No smallness assumption on δ or u is required.

Proof. The error $e = H - \widehat{H}$ satisfies

$$D_C^\alpha e = \left(d + \frac{H + \widehat{H}}{2}\right)e - r, \quad e(0) = 0, \quad \operatorname{Re}\left(d + \frac{H + \widehat{H}}{2}\right) \le -\sigma.$$

Apply (C.4) to $\psi_\epsilon(e)$ to obtain $D_C^\alpha \psi_\epsilon(e) + \sigma \psi_\epsilon(e) \le |r| \le \delta$. Comparison with the linear scalar solution, followed by $\epsilon \downarrow 0$, proves the claim. Convex regularisation includes points of zero error and applies to the complex error viewed as a vector in the real plane. ∎

If $\widehat{H}$ corresponds to the physical-time trajectory $\widehat{Z}$ and $|r_t|/\nu \le \delta_F$ has been certified, the same result gives

$$\sup_{t\le T}|Z-\widehat{Z}|\le \delta_F/s_0 \quad \text{if } \operatorname{Re}\widehat{Z}\le 0. \tag{C.8}$$

In the numerical example, $s_0 = 1489/4000$. This dissipation rate controls the complex error modulus; Appendix D.1 supplies the independent residual δ_F. The estimate applies the established convexity tool to the specific error propagation considered here.

Li and Liu [LiLiu2018, Proposition 3.11(ii)] give Caputo convexity by regularization and a distributional limit; their Proposition 4.12 treats specified vector gradient and Hamiltonian flows. Kopteva [Kopteva2021, Lemma 2.8 and Theorem 2.2] uses norm convexity and a positive fractional inverse under positive-time Lipschitz hypotheses. The following lemma proves the vector AC form needed here, including the terminal-time conclusion.

Regularity lemma (AC convexity and the continuous endpoint). Let $0<\alpha<1$, $T>0$, $v\in AC([0,T];\mathbb{R}^d)$, and let $\Phi\in C^1(\mathbb{R}^d)$ be convex. With $D_C^\alpha v = g_{1-\alpha}*v'$,

$$D_C^\alpha \Phi(v)\le \nabla\Phi(v)\cdot D_C^\alpha v \quad \text{a.e. on } (0,T). \tag{C.9}$$

Let $u\in AC[0,T]$, $u(0)=0$, $\lambda\ge 0$, and $R\in L^\infty(0,T)$ be nonnegative. If $D_C^\alpha u+\lambda u\le R$ almost everywhere, then

$$u(t)\le (k_\lambda * R)(t), \qquad 0\le t\le T, \quad k_\lambda(t)=t^{\alpha-1}E_{\alpha,\alpha}(-\lambda t^\alpha). \tag{C.10}$$

Proof. Choose smooth $f_n\to v'$ in $L^1(0,T)$, and set $v_n(t)=v(0)+\int_0^t f_n(s)\,ds$. Then $v_n(0)=v(0)$, $v_n\to v$ uniformly, and $v_n'\to v'$ in L^1. For a smooth path, integration by parts expresses the difference between the two sides of (C.9) as

$$\frac{1}{\Gamma(1-\alpha)}\left[\frac{B_\Phi(v_n(0),v_n(t))}{t^\alpha}+\alpha\int_0^t \frac{B_\Phi(v_n(s),v_n(t))}{(t-s)^{1+\alpha}}\,ds\right]\ge 0, \tag{C.11}$$

where $B_\Phi(a,b)=\Phi(a)-\Phi(b)-\nabla\Phi(b)\cdot(a-b)\ge 0$. Smooth paths are locally Lipschitz. A gradient bound M on the compact path range and a local path Lipschitz constant L give $B_\Phi(v_n(s),v_n(t))\le 2ML|t-s|$. The endpoint kernel $(t-s)^{-\alpha}$ is therefore integrable. The supporting-hyperplane inequality may first be integrated with history truncated at $t-\delta$, and the classical Caputo integrals converge as $\delta\downarrow 0$.

All path ranges lie in one compact set. Continuity of $\nabla\Phi$ and the ordinary AC composition rule give

$$\|\Phi(v_n)'-\Phi(v)'\|_1\to 0, \qquad \|D_C^\alpha(v_n-v)\|_1\le \frac{T^{1-\alpha}}{\Gamma(2-\alpha)}\|v_n'-v'\|_1\to 0. \tag{C.12}$$

The same convolution estimate applies to $\Phi(v_n)-\Phi(v)$. Moreover, the products $\nabla\Phi(v_n)\cdot D_C^\alpha v_n$ converge in L^1 to the corresponding product for v. Passing to an almost-everywhere convergent subsequence proves (C.9). This uses the ordinary composition rule only for the first derivative inside AC; no Caputo chain rule is asserted.

For (C.10), put $f=D_C^\alpha u+\lambda u\in L^1(0,T)$. The zero initial value gives $u+\lambda g_\alpha * u = g_\alpha * f$. The locally convergent resolvent series, whose n-th term has $L^1(0,T)$ norm bounded by $\lambda^{n-1}T^{n\alpha}/\Gamma(1+n\alpha)$, gives $u=k_\lambda * f$ almost everywhere. Complete monotonicity of $E_{\alpha,\alpha}(-x)$ [SimonCM2015] supplies $k_\lambda\ge 0$, so $f\le R$ implies $u\le k_\lambda * R$ almost everywhere. The convolution on the right is continuous: extend k_λ and R by zero, and use L^1 translation continuity of the kernel with the L^∞ bound on R. Its value at zero is zero. Since u is continuous, an almost-everywhere inequality extends to every point of $[0,T]$, including maturity. ∎

C.2. Proof of Theorem 3.1

Proof. For $e = Z - \widehat{Z}$, the divided difference gives $D_C^\alpha e = \nu[d + (Z + \widehat{Z})/2]e - r$, with coefficient real part at most $-\lambda$. Apply the regularity lemma to $v_\varepsilon = (|e|^2 + \varepsilon^2)^{1/2} - \varepsilon$. Since $\nabla v_\varepsilon \cdot e = |e|^2/(|e|^2 + \varepsilon^2)^{1/2} \geq v_\varepsilon$ and $|\nabla v_\varepsilon| \leq 1$,

$$D_C^\alpha v_\varepsilon + \lambda v_\varepsilon \leq R \quad \text{a.e.} \tag{C.13}$$

Equation (C.10) and the limit $\varepsilon \downarrow 0$ yield the first state inequality at every time. Absolute Fubini gives $I^{1-\alpha}\xi = V_0 g_{2-\alpha} + g_{2-\alpha} * \xi'$, whose derivative is (3.4), and $g_\alpha * q_\alpha = \xi$. In particular, $q_\alpha \in L^1$ and $\xi \geq 0$. Writing $A_\alpha = I^{1-\alpha}\xi$, another absolute Fubini step followed by AC integration by parts gives

$$L_T - \widehat{L}_T = \nu^{-1} \int_0^T A_\alpha(T-t) e'(t)\, dt = \nu^{-1}(q_\alpha * e)(T). \tag{C.14}$$

The first Fubini integral is bounded by $\|\xi\|_\infty T^{1-\alpha}\|e'\|_1/\Gamma(2-\alpha)$; the boundary terms vanish because $A_\alpha(0) = e(0) = 0$. Positivity now proves the first exponent estimate. Finally, $g_\alpha - k_\lambda = \lambda g_\alpha * k_\lambda \geq 0$, whence

$$0 \leq q_\alpha * k_\lambda = \xi - \lambda \xi * k_\lambda \leq \xi. \tag{C.15}$$

Associativity and nonnegative integration yield the remaining conclusions. The physical residual is R, so its factor ν^{-1} remains until $R/\nu \leq \delta_F$ is used. ∎

The proof also covers $\lambda = 0$: the resolvent is g_α, every convolution remains integrable, and no division by σ occurs.

C.3. Curve positivity and explicit constant-curve bounds

The curve condition in Theorem 3.1 is the nonnegativity of $q_\alpha = (I^{1-\alpha}\xi)'$, rather than monotonicity of ξ. The state and reference still belong to $AC[0,T]$, have the same zero initial value, and satisfy the full-time residual and dissipation hypotheses in (3.3). The physical residual remains R, the normalized residual is R/ν, and physical damping is $\lambda = \nu\sigma > 0$. Both exponents retain the D-type definitions in Section 3. These assumptions are required for each application; the condition on the pricing kernel does not prove stochastic-model existence or extend the verified Riccati regularity domain.

For a real curve $\xi \in AC[0,T]$ with $\xi(0) = V_0 \geq 0$, the precise sufficient condition is

$$q_\alpha = V_0 g_{1-\alpha} + g_{1-\alpha} * \xi' \geq 0 \quad \text{a.e.}, \qquad g_\beta(t) = t^{\beta-1}/\Gamma(\beta). \tag{C.16}$$

No sign is imposed on ξ'. Indeed, $A_\alpha = I^{1-\alpha}\xi = V_0 g_{2-\alpha} + g_{2-\alpha} * \xi'$ is absolutely continuous, $A_\alpha(0) = 0$, and

$$\|q_\alpha\|_1 \leq \frac{T^{1-\alpha}}{\Gamma(2-\alpha)}(V_0 + \|\xi'\|_1), \qquad g_\alpha * q_\alpha = \xi. \tag{C.17}$$

Absolute Fubini proves the second identity even for signed ξ', using $g_\alpha * g_{1-\alpha} = g_1$. The initial term $V_0 g_{1-\alpha}$ cannot be dropped. Condition (C.16) implies $\xi \geq 0$. Hence the entire proof of Theorem 3.1 continues to hold: integration by parts gives $L_T - \widehat{L}_T = \nu^{-1}(q_\alpha * e)(T)$, and the positive resolvent identity gives

$$0 \leq q_\alpha * k_\lambda = \xi - \lambda \xi * k_\lambda \leq \xi. \tag{C.18}$$

The Fubini step for the exponent is bounded by $\|\xi\|_\infty T^{1-\alpha}\|e'\|_1/\Gamma(2-\alpha)$; its boundary terms vanish because $e(0) = A_\alpha(0) = 0$. The AC Caputo convexity and the positive

zero-initial-value inverse remain the prior tools [LiLiu2018, Kopteva2021]. The finite-history convolution, cell-weight bound (3.7), and outward node-radius construction therefore remain valid under (C.16). Cells still carry the complete Caputo history.

This is a strict relaxation of a sufficient curve condition, not a necessary condition for every conceivable certificate. If q_α changes sign, the safe general estimate is $\nu^{-1}(|q_\alpha| * k_\lambda * R)(T)$; the positive collapse to $\xi * R$ cannot be used without further proof. Fractional forward-variance representations and their model-admissibility restrictions already occur in [ElEuchRosenbaum2017, Proposition 3.1 and Corollary 3.3]. Our analytical condition does not replace those restrictions.

Corollary C.3 (a decreasing analytical curve). *Let* $V_0 > 0$, $c > 0$, *and* $\xi(t) = V_0(1 - ct)$. *Then*

$$q_\alpha(t) = \frac{V_0 t^{-\alpha}}{\Gamma(1-\alpha)}\left(1 - \frac{ct}{1-\alpha}\right). \tag{C.19}$$

Thus (C.16) holds on $(0, T)$ if and only if $cT \le 1 - \alpha$, including equality. In that case $\xi(T) \ge \alpha V_0 > 0$ despite $\xi' < 0$.

Proof. Convolving the constant derivative $-V_0 c$ with $g_{1-\alpha}$ gives $-V_0 c\, t^{1-\alpha}/\Gamma(2-\alpha)$. Adding the initial term and using $\Gamma(2-\alpha) = (1-\alpha)\Gamma(1-\alpha)$ proves (C.19). Its bracket is nonnegative exactly under the stated condition. ∎

The example illustrates the analytical propagation condition. Realizing it as a rough-Heston variance curve would require a separate model-admissibility proof. Positivity of the curve alone is insufficient for (C.16): $1 - \alpha < cT < 1$ leaves $\xi > 0$ but gives a negative kernel near maturity.

Corollary C.4 (explicit constant-curve propagation). *Under Theorem 3.1's state hypotheses with* $\sigma > 0$, *let* $\xi \equiv V_0 \ge 0$. *Then*

$$q_\alpha * k_\lambda = V_0 E_\alpha(-\lambda t^\alpha), \qquad |L_T - \widehat{L}_T| \le \frac{V_0}{\nu}\int_0^T E_\alpha(-\lambda(T-s)^\alpha) R(s)\, ds. \tag{C.20}$$

If $R \le \nu\delta_F$, a closed-form upper bound is

$$\eta_{\mathrm{ML}} = V_0 \delta_F T E_{\alpha,2}(-\nu\sigma T^\alpha) \le \min\left\{V_0\delta_F T, \frac{V_0 \delta_F T^{1-\alpha}}{\nu\sigma\Gamma(2-\alpha)}\right\}. \tag{C.21}$$

Proof. Termwise convolution in the absolutely locally integrable Mittag–Leffler series gives $g_{1-\alpha} * k_\lambda = E_\alpha(-\lambda t^\alpha)$; termwise integration gives $\int_0^T E_\alpha(-\lambda t^\alpha)dt = T E_{\alpha,2}(-\lambda T^\alpha)$. Equation (C.18) gives $0 \le E_\alpha \le 1$, proving the first comparison. Also $\int_0^t k_\lambda = (1 - E_\alpha(-\lambda t^\alpha))/\lambda \le 1/\lambda$. The state envelope $|e| \le \delta_F/\sigma$, followed by the positive q_α integral, proves the second comparison. ∎

The explicit bound retains finite history and physical scaling. It may be intersected with other valid exponent bounds for the same derivative-based target and reference. A substitution-based reference requires the certified residual conversion. The corollary gives an analytical constant-curve comparison; Section 7.5 computes the resolvent kernel for the fixed nonconstant experimental curve.

Proposition C.5 (an error radius without an assumed approximate half-plane). *Assume* $s_0 > 0$, $0 \le \delta_F < s_0^2/2$, $D_C^\alpha Z = \nu F(Z)$, $\Re Z \le 0$, *and* $|D_C^\alpha \widehat{Z} - \nu F(\widehat{Z})| \le \nu\delta_F$ *almost everywhere. The exact and approximate trajectories are AC, locally Lipschitz at positive times, and have the same zero initial value. Then*

$$|Z-\widehat{Z}| \le E_\delta = s_0 - \sqrt{s_0^2 - 2\delta_F} = \frac{2\delta_F}{s_0 + \sqrt{s_0^2 - 2\delta_F}}. \tag{C.22}$$

To prove this, rewrite the error terms as $(d+Z)e - e^2/2$. The exact-solution half-plane gives $D_t^\alpha |e| \le \nu(-s_0|e| + |e|^2/2 + \delta_F)$, interpreted through the same convex regularisation of the modulus. Choose any $E_\delta < \ell < s_0 + \sqrt{s_0^2 - 2\delta_F}$. The right-hand side is strictly negative at $|e| = \ell$; a first hitting time of ℓ contradicts (C.3). Letting $\ell \downarrow E_\delta$ proves the result. This is a sufficient small-residual condition; violating it does not establish an actual error or pole. The completed node trajectories in this example have a certified half-plane property and use (C.8), which does not impose the threshold in (C.22).

C.4. Complementary global exponent bound

For (2.4), $\xi_* \in AC$, $\xi_*' \ge 0$, and $V_0 \le \xi_* \le \theta$. Positivity of the scalar resolvent follows from the negative-minimum principle in (C.3), or from the completely monotone representation of Simon [SimonCM2015]. Define

$$A_\alpha(t) = (I^{1-\alpha}\xi_*)(t), \quad q_\alpha(t) = A_\alpha'(t) = \frac{V_0 t^{-\alpha}}{\Gamma(1-\alpha)} + I^{1-\alpha}\xi_*'(t) \ge 0. \tag{C.23}$$

Both initial exponents $-\alpha, \alpha_0 - \alpha$ exceed -1, so $q_\alpha \in L^1(0,T)$. Explicitly,

$$q_\alpha(t) = \frac{V_0 t^{-\alpha}}{\Gamma(1-\alpha)} + (\theta - V_0)\lambda_\xi t^{\alpha_0-\alpha} E_{\alpha_0, 1+\alpha_0-\alpha}(-\lambda_\xi t^{\alpha_0}), \tag{C.24}$$

$$A_\alpha(t) = \frac{\theta t^{1-\alpha}}{\Gamma(2-\alpha)} + (V_0 - \theta) t^{1-\alpha} E_{\alpha_0, 2-\alpha}(-\lambda_\xi t^{\alpha_0}), \quad A_\alpha(0) = 0. \tag{C.25}$$

where $E_{a,b}(z) = \sum_{n=0}^\infty z^n/\Gamma(an+b)$.

Theorem C.6. *For a zero-initial-value AC trajectory $Z, \widehat{Z}$, use the derivative-based exponent*

$$L_T = \nu^{-1}\int_0^T \xi_*(T-t) D_t^\alpha Z(t)\,dt, \quad \bar{L}_T = \nu^{-1}\int_0^T \xi_*(T-t) D_t^\alpha \widehat{Z}(t)\,dt,$$

Then

$$L_T - \bar{L}_T = \nu^{-1}\int_0^T q_\alpha(T-t)(Z-\widehat{Z})(t)\,dt. \tag{C.26}$$

In particular, if $\sup|Z-\widehat{Z}| \le E$, then

$$|L_T - \bar{L}_T| \le \eta = E A_\alpha(T)/\nu \le \frac{E\theta T^{1-\alpha}}{\nu\Gamma(2-\alpha)}. \tag{C.27}$$

Proof. Since $D_t^\alpha v = I^{1-\alpha} v'$, absolute Fubini changes the exponent integral into $\nu^{-1}\int_0^T A_\alpha(T-t)v'(t)\,dt$. Integration by parts, using $A_\alpha(0) = v(0) = 0$, yields the positive-kernel representation. Absolute Fubini is justified by $v' \in L^1$ and the continuous, finite curve. Then $q_\alpha \ge 0$ and $\int q_\alpha = A_\alpha(T)$ give the estimate. ∎

If $\widehat{Z} = I^\alpha \bar{G}$, then $\bar{L} = \nu^{-1}\int \xi_* \bar{G}$ is exactly this derivative-based exponent, so the residual integral is not added again. The substitution-based exponent formed from $\int \xi_* F(\widehat{Z})$ differs by $\nu^{-1}\int \xi_* r_t$ and requires a separate conversion. All numerical exponent quantities used here are derivative-based.

Using $\operatorname{Re} L_T \le 0$ and the integral identity for the exponential, whenever (C.27) holds,

$$|e^{L_T} - e^{\bar{L}_T}| \le \min\{1, e^{\operatorname{Re} \bar{L}_T}\}(e^{\eta} - 1). \tag{C.28}$$

The first estimate is based at the exact exponent; the second is based at the approximate exponent. Their minimum is a valid bound. The state error may increase with frequency, while decay of the approximate transform under the pricing weight can compensate for part of that increase.

If a model-transform envelope $B(u)$ and an approximate-modulus bound $\widehat{B}(u)$ are also available, then

$$|e^{L_T} - e^{\bar{L}_T}| \le \min\{B + \widehat{B},\ \eta(B + \widehat{B})/2\}. \tag{C.29}$$

The first bound is the triangle inequality. The second follows from $(L_T - \bar{L}_T)\int_0^1 e^{(1-\tau)\bar{L}_T + \tau L_T} d\tau$ and the convexity bound $e^{(1-\tau)a+\tau b} \le (1-\tau)e^a + \tau e^b$. Since (C.28) and (C.29) refer to the same approximate transform, their radii may be minimized. Setting the reference to zero instead gives a disk of radius $B(u)$ about a different centre. It may be compared only after translating the centres as in Proposition 5.2, or by intersecting the complete price intervals. Radii about distinct centres are not directly minimized.

C.5. Certified dissipative-kernel weights

Theorem 3.1 already includes zero damping; specialize its full assumptions and AC comparison proof to $\sigma = 0$. The regularized error modulus satisfies $D_C^\alpha v_\varepsilon + \nu\sigma v_\varepsilon \le R$ almost everywhere. The same positive inverse applies when $\lambda = \nu\sigma = 0$: $k_0 = g_\alpha$. Passing to the continuous endpoint and then composing with the pricing functional gives (3.8)–(3.9). In particular the zero-damping assertion introduces no $1/\sigma$. The initial contribution $V_0 g_{1-\alpha}$ in q_α and the factor ν^{-1} are unchanged.

Proposition (a rigorous cumulative-weight series). *Suppose* $0 < \alpha, A_0 < 1$, $0 \le V_0 \le \theta$, *and the fixed curve is* $\xi(t) = \theta + (V_0 - \theta)E_{A_0}(-\lambda_\xi t^{A_0})$, *with* $\lambda_\xi \ge 0$. *Retain* $q_\alpha \ge 0$. *For* $\lambda \ge 0$ *and* $\lambda T^\alpha < 1$, *(3.10) holds on* $[0, T]$. *If it is truncated before index* M, *the entire omitted outer series satisfies*

$$\left| W_\lambda(t) - \sum_{n=0}^{M-1} (-\lambda)^n W_n(t) \right| \le \frac{\theta t (\lambda t^\alpha)^M}{1 - \lambda t^\alpha}, \quad W_n(t) = \int_0^t (\xi * g_{n\alpha})(s)\, ds. \tag{C.30}$$

Here g_0 denotes the identity convolution, so $W_0(t) = \int_0^t \xi$.

Proof. The existing resolvent identity $k_\lambda + \lambda g_\alpha * k_\lambda = g_\alpha$, together with $q_\alpha * g_\alpha = \xi$, gives $K_\lambda + \lambda g_\alpha * K_\lambda = \xi$. The Neumann series has terms $(-\lambda)^n(\xi * g_{n\alpha})$. Since $0 \le \xi \le \theta$,

$$0 \le W_n(t) \le \frac{\theta t^{1+n\alpha}}{\Gamma(2 + n\alpha)} \le \theta t (t^\alpha)^n. \tag{C.31}$$

Log convexity and $\Gamma(1) = \Gamma(2) = 1$ imply that Γ is increasing and at least one on $[2, \infty)$. Every outer denominator argument is $2 + n\alpha \ge 2$. Thus the sum of the $L^1(0, T)$ norms of the absolute terms is bounded by the convergent geometric series in (C.31). This justifies convolution, integration and interchange, proves the Volterra identity and its unique resolvent solution, and gives (C.30) with the first omitted index exactly $n = M$. In particular twelve retained terms omit $n \ge 12$, rather than $n \ge 13$.

For uniqueness, the difference $h \in L^1(0, T)$ of two solutions satisfies $h = -\lambda g_\alpha * h$. Iterating gives $\|h\|_1 \le \lambda^n T^{n\alpha} \|h\|_1 / \Gamma(1 + n\alpha)$. For sufficiently large n, the Gamma argument is at least two, and the right-hand factor is at most $(\lambda T^\alpha)^n$, which tends to zero.

Thus $h = 0$. This argument does not assume an unjustified one-step contraction involving $\Gamma(1+\alpha)$.

Expanding the curve Mittag–Leffler function, and using the beta convolution identity, gives

$$W_n(t) = t^{1+n\alpha}\left[\frac{\theta}{\Gamma(2+n\alpha)} + (V_0-\theta)\sum_{m=0}^{\infty}\frac{(-\lambda_\xi t^{A_0})^m}{\Gamma(2+n\alpha+mA_0)}\right]. \tag{C.32}$$

When $w = \lambda_\xi t^{A_0} < 1$, every inner denominator also has argument at least two, and its tail before index L is bounded by

$$\left|\sum_{m=L}^{\infty}\frac{(-w)^m}{\Gamma(2+n\alpha+mA_0)}\right| \le \frac{w^L}{1-w}. \tag{C.33}$$

The tails are absolute enclosures; no unproved alternating-series monotonicity is needed. The actual curve and damping satisfy the strict interval checks $w < 1$ and $\lambda T^\alpha < 1$. Signed coefficients, including $V_0 - \theta$, are multiplied by outward intervals before addition. Every power, Gamma value and exponential is enclosed by the identified dyadic primitives. ∎

For a residual envelope $R \le R_j$ on the complete cell $[a_j, b_j]$, evaluate strict cumulative intervals $[W^-(t), W^+(t)]$. Its exact weight lies in

$$\big[W^-(T-a_j) - W^+(T-b_j),\ W^+(T-a_j) - W^-(T-b_j)\big]. \tag{C.34}$$

Intersecting with nonnegativity and the independently proved upper curve weight is valid because $0 \le K_\lambda \le \xi$. The resulting upper endpoint ω_j^+ gives

$$\eta_{\mathrm{res}} \le \nu^{-1}\sum_j R_j \omega_j^+. \tag{C.35}$$

Every bin envelope is the maximum over all intersecting closed source cells, including endpoint intersections. Source cells still start at zero and cover the full interval; no Caputo history is restarted.

The producer evaluates W_n through the strict power-field moment divided by $\Gamma(1+n\alpha)$, retaining twelve outer terms. The separate reader evaluates (C.32) directly, retaining fourteen outer terms and sixty-four inner terms, with its own cumulative and moment functions. It checks that the reported residual-functional enclosure contains its tighter interval, then checks the transform-radius minima and complete prices. The two calculations share Gamma, logarithm, exponential and dyadic coefficient primitives and use the same upstream residual proof.

Appendix D. Continuous certification and reliable refinement

D.1. Reference trajectory, startup and arithmetic

For each candidate $\alpha = \beta \in \{.52, .6, .9\}$ and each $u = n/8$, $n = 0, \dots, 512$, we store

$$\bar G(t) = \nu c_0 + A_1 t^\beta + A_2 t^{2\beta} + \bar L(t), \quad c_0 = -(u^2+1/4)/2, \quad \widehat Z = I^\alpha \bar G, \quad T = 1/2. \tag{D.1}$$

Here $\bar L$ is the continuous piecewise-linear interpolation in physical time at nodes (t_j, L_j) , with $L_0 = 0$. Nodes and complex coefficients are interpreted as the exact dyadic rationals represented by their binary64 values. The following independent residual controls the continuous-time error of this reference trajectory:

$$r_t = \bar{G} - \nu F(\widehat{Z}), \quad \delta_F = \nu^{-1} \sup_{0 \le t \le T} |r_t|. \tag{D.2}$$

Equation (D.1) defines an entire absolutely continuous reference trajectory. Its independent residual gives the model-price intervals, which are then used to bound the output error of the selected [3/3] implementation.

Write

$$\widehat{Z} = H_0 + J, \quad J = I^{\alpha}\bar{L}, \quad H_0 = B_1 t^{\alpha} + B_2 t^{\alpha+\beta} + B_3 t^{\alpha+2\beta},$$

$$B_1 = \frac{\nu c_0}{\Gamma(1+\alpha)}, \quad B_2 = \frac{A_1 \Gamma(1+\beta)}{\Gamma(1+\alpha+\beta)}, \quad B_3 = \frac{A_2 \Gamma(1+2\beta)}{\Gamma(1+\alpha+2\beta)}. \tag{D.3}$$

The initial interval is $J = L_1 t^{\alpha+1}/[t_1 \Gamma(2+\alpha)]$. Substitute all generalised powers into (D.2), first cancel νc_0 exactly, combine equal powers, and then take $\sum_p |r_p| t_1^p$. Since every $p > 0$, this bounds the full initial closed interval. The generalised-power expansion retains the fractional initial behaviour.

On a source interval $[a, b] \subset [0, q]$, set $\tau = q - a, h = b - a$. The integral weights for the linear hat functions are

$$I_0 = \{\tau^{\alpha} - (\tau - h)^{\alpha}\}/\alpha, \quad I_1 = \{\tau^{\alpha+1} - (\tau - h)^{\alpha+1}\}/(\alpha+1),$$

$$w_R = (\tau I_0 - I_1)/(h\Gamma(\alpha)), \quad w_L = I_0/\Gamma(\alpha) - w_R. \tag{D.4}$$

These weights are nonnegative. For a truncated interval, recover the endpoints by affine interpolation within the original interval. On the final interval, $b = q$ can be evaluated directly using $w_R = h^{\alpha}/[\alpha(\alpha+1)\Gamma(\alpha)], w_L = \alpha w_R$. For $h/\tau < .01$, retain eight positive-series terms, using respectively

$$w_R = \frac{h\tau^{\alpha-1}}{\Gamma(\alpha)} \sum_{k \ge 0} \frac{(1-\alpha)_k (h/\tau)^k}{k!(k+2)}, \quad w_L = \frac{h\tau^{\alpha-1}}{\Gamma(\alpha)} \sum_{k \ge 0} \frac{(1-\alpha)_k (h/\tau)^k}{k!(k+1)(k+2)}. \tag{D.5}$$

The unscaled tails are each bounded by $z^8/(1-z)$ because $0 < (1-\alpha)_k/k! \le 1$. This reduces interval inflation from subtracting nearly equal endpoint quantities while retaining rigorous history integration.

Partition each nonstartup source interval into closed subintervals $[a, b]$: H/Q banks use four subintervals, and nearby-candidate banks use two. Choose the recorded dyadic centre m of each subinterval. Given a derivative bound on that full subinterval,

$$\sup_{[a,b]} |r_t| \le |r_t(m)| + \max(m - a, b - m) \sup_{[a,b]} |r_t'|. \tag{D.6}$$

The ordinary residual derivative may be discontinuous at original nodes. The derivative bounds below hold almost everywhere; absolute continuity of the residual and the fundamental theorem of calculus still imply (D.6). Each of the following three identities provides a derivative bound, and we take their minimum:

$$r_t' = \bar{G}' - \nu(d + \widehat{Z})\widehat{Z}',$$

$$P_r = \nu c_0 + A_1 t^{\beta} + A_2 t^{2\beta} - \nu F(H_0), \quad r_t = P_r + \bar{L} - \nu(d + H_0)J - \nu J^2/2,$$

$$r_t' = P_r' + \bar{L}' - \nu[(d + H_0 + J)J' + H_0' J], \quad \widehat{Z}' = \frac{\nu c_0\, t^{\alpha-1}}{\Gamma(\alpha)} + I^{\alpha}\bar{G}'. \tag{D.7}$$

The constant-kernel weight of a past source interval decreases with the integration endpoint, whereas the weight of the current source interval increases while that endpoint lies within it. Endpoint bounds therefore enclose both types of weight. In the third identity, first combine $\bar{G}'(s) = \beta A_1 s^{\beta-1} + 2\beta A_2 s^{2\beta-1} + \bar{L}'(s)$ on each source interval and then integrate against the positive kernel, retaining cancellation in the original relation.

The singular terms from the initial source interval are handled directly by fractional-power integrals. For $p > 0, t_1 \le a \le q \le b$, the positive weight satisfies

$$\frac{t_1^p b^{\alpha-1}}{p\Gamma(\alpha)} \le \frac{1}{\Gamma(\alpha)} \int_0^{t_1} s^{p-1}(q-s)^{\alpha-1}\,ds \le \min\left\{ \frac{t_1^p (a-t_1)^{\alpha-1}}{p\Gamma(\alpha)}, \frac{\Gamma(p)}{\Gamma(\alpha+p)} \max_{q\in[a,b]} q^{\alpha+p-1} \right\}. \tag{D.8}$$

When $a = t_1$, discard the first upper bound. The second follows from the full Euler Beta integral and is finite. Kernel monotonicity at the endpoints gives the lower bound and the first upper bound. Every $p, \alpha > 0$, so the hypotheses of the classical Beta formula hold.

Using the same $\widehat{Z}'$ enclosure and rigorous point values, certify an upper bound for $\operatorname{Re}\widehat{Z}$ on every subsequent subinterval. On the initial interval, factor out t^α, leaving a finite bracket; use its negative constant term and the endpoint contributions of the remaining positive parts. Bound (C.8) is used only when the upper real-part bound is nonpositive on every closed interval. This condition therefore covers the entire closed time interval.

The structural certificate and special functions use outward-rounded intervals with integer endpoints divided by 2^{100}. Batched residual weights use IEEE binary64; each basic operation is enlarged outward to adjacent representable numbers. After converting an exact rational Gamma enclosure, endpoint inclusion is checked with Fraction. The logarithm uses exact binary scaling and a twenty-two-term atanh series with $z \in [0, 1/3]$, whose tail is $2z^{45}/[45(1-z^2)]$. After scaling, the exponential has $|r| < 1$ and a degree-24 Taylor tail bounded by $3/25!$. These explicit remainders define the intervals for logarithms, exponentials, and noninteger powers.

For matrix multiplication with rigorous weight centres W_c, radii W_r, and recorded dyadic values X, include

$$(\gamma_{2n}\|W_c\|_{1,\mathrm{row}} + \|W_r\|_{1,\mathrm{row}})\|X\|_{\infty,\mathrm{column}}, \quad \gamma_{2n} = \frac{2n\,2^{-53}}{1 - 2n\,2^{-53}}, \tag{D.9}$$

and an additional underflow-error allowance. Induction on products of basic rounding factors gives this estimate and permits different summation association orders. Norm sums are also rounded outward. The calculation assumes round-to-nearest IEEE basic operations, correctly rounded square roots, gradual underflow, and no overflow or NaNs. Appendix E identifies the implementation and arithmetic dependencies; interval inclusion uses these assumptions and the stated remainder bounds.

For the fixed curve, define

$$J_0(z) = \theta z + (V_0 - \theta) z E_{\alpha_0,2}(-\lambda_\xi z^{\alpha_0}), \quad J_1(z) = \theta z^2/2 + (V_0 - \theta) z^2 (E_{\alpha_0,2} - E_{\alpha_0,3})(-\lambda_\xi z^{\alpha_0}). \tag{D.10}$$

These quantities are respectively $\int_0^z \xi_*(s)ds, \int_0^z s\xi_*(s)ds$; termwise integration verifies the second formula. If a linear-trajectory interval is $[a, b]$ with $A = T - b, B = T - a$, its rigorous nonnegative exponent weights are

$$w_l = \frac{J_1(B) - J_1(A) - A[J_0(B) - J_0(A)]}{b-a}, \quad w_r = \frac{B[J_0(B) - J_0(A)] - J_1(B) + J_1(A)}{b-a}. \tag{D.11}$$

The curve moment for the power component t^β is

$$\int_0^T \xi_*(T-t)t^\beta dt = \frac{\theta T^{\beta+1}}{\beta+1} + (V_0-\theta)\Gamma(\beta+1)T^{\beta+1}E_{\alpha_0,\beta+2}(-\lambda_\xi T^{\alpha_0}). \tag{D.12}$$

This follows directly from the series and Beta integration. Together with (D.1), it makes $\bar{L}$ a rigorous finite sum. Trigonometric phases and complex exponentials also use finite Taylor expansions with explicit tails. On the present horizon, if $z = \lambda_\xi T^{\alpha_0} < 1$, a Mittag–Leffler series tail may be bounded by $3z^{64}/(1-z)$: all Gamma arguments exceed one, and Euler's integral gives $\Gamma(\gamma) \geq e^{-1} > 1/3$. Rigorous integration encloses the approximate exponent, while (C.27) controls the difference between the model and approximate exponents.

D.2. Strip, exact high-frequency tail and reference field

Proof. Jensen's inequality gives, within the strip, $|\phi_T(z-i/2)| \leq \mathbb{E}M_T^{1/2-\operatorname{Im} z} \leq 1$. On compact substrips, $x^q|\log x|^j \leq C(1+x)$, so the transform is analytic. The boundaries $z = r \pm ia_*$ satisfy

$$|z^2+1/4| \geq r^2 + (1/2-a_*)^2, \quad \int_{\mathbb{R}} |g(r \pm ia_*)|\,dr \leq M_* = \frac{\pi e^{a_*|k|}}{1/2-a_*}.$$

Shift the real-axis Fourier integration contour to the upper and lower boundaries. The vertical integrals vanish by quadratic decay, giving $|\tilde{g}(\omega)| \leq M_* e^{-a_*|\omega|}$. The absolutely and uniformly convergent periodisation $\sum_{j\in\mathbb{Z}} g(x+jh_*)$ has Fourier coefficients $h_*^{-1}\tilde{g}(2\pi n/h_*)$. Summing at zero and subtracting $n=0$ gives a two-sided error of at most $2M_*/(e^{2\pi a_*/h_*}-1)$. Conjugate symmetry halves both the two-sided integral and its sum. Multiplication by $\sqrt{m}/\pi$ gives (4.4). ∎

This is a specific application of the analytic-strip trapezoidal theory of Trefethen and Weideman [TrefethenWeideman2014]. Each pricing node uses its own continuous-time residual bound; the analytic strip controls the quadrature error between nodes.

Fix $\kappa = 0, -1 < \rho < 0$. Set $X = -\operatorname{Re} H \geq 0$. Equation (C.6) gives

$$D_x^\alpha X \geq \beta_u - s_0 X - X^2/2, \quad \beta_u = (1-\rho^2)u^2/2 + 1/8. \tag{D.13}$$

Let w be the scalar zero-initial-value solution satisfying equality. Comparison gives

$$X \geq w, \quad 0 \leq w \leq R_u, \quad R_u = \sqrt{s_0^2 + 2\beta_u} - s_0, \quad \ell_u = (s_0 + \sqrt{s_0^2+2\beta_u})/2.$$

Because the right-hand side satisfies $(R_u - w)(s_0 + (R_u+w)/2) \geq \ell_u(R_u - w)$, linear comparison and the bound of Simon [Simon2014], $E_\alpha(-z) \leq (1 + z/\Gamma(1+\alpha))^{-1}$, give

$$-\operatorname{Re} Z(t,u) \geq R_u \frac{\ell_u \nu t^\alpha}{\Gamma(1+\alpha) + \ell_u \nu t^\alpha}. \tag{D.14}$$

The comparison argument uses a positive maximum over the full history.

Take $V > s_0/\sqrt{1-\rho^2}$ and define

$$a_\rho = \sqrt{1-\rho^2}, \quad b_V = a_\rho - s_0/V > 0, \quad f_\alpha(z) = z/(\Gamma(1+\alpha)+z).$$

For $u \geq V$, we have $R_u/u \geq b_V, \ell_u \geq a_\rho u/2$. For any partition of $J \geq 2$, denoted $0 = t_0 < \cdots < t_J = T$, the positive kernel in (C.26) and the monotonicity of f_α give

$$c_V = \frac{b_V}{\nu} \sum_{j=0}^{J-1} f_\alpha(a_\rho \nu V t_j^\alpha/2)[A_\alpha(T-t_j) - A_\alpha(T-t_{j+1})] > 0,$$

$$|\phi_T(u - i/2)| \le e^{-c_V u} \quad (u \ge V). \tag{D.15}$$

This envelope holds at every continuous high frequency. The time partition constructs a lower sum for the positive-measure integral. The example uses $J = 64$ and rigorous intervals to obtain a positive rational lower bound for c_V. No singular evaluation is performed at $A_\alpha(0) = 0$. The difference between bounds taken from the appropriate sides at distinct endpoints, $[A_{\mathrm{lo}}(T-t_j) - A_{\mathrm{hi}}(T-t_{j+1})]_+$, is a valid lower bound on the mass.

Using the right-endpoint sum of a positive decreasing function, for $V = N_u h_*$ the exact discrete-tail contribution to the normalised price is bounded by

$$\epsilon_{\mathrm{tail}} \le \frac{\sqrt{m}}{\pi} \frac{e^{-c_V V}}{c_V V^2}. \tag{D.16}$$

The integral tail uses an envelope for the exact solution. Finite nodes assigned the value zero are charged according to the same envelope, so every part of the frequency range has an explicit error contribution.

For each candidate, a separately implemented product-integration procedure generates and fixes a physical-time reference function $\bar{G} = \nu c_0 + A_1 t^\beta + A_2 t^{2\beta} + \bar{L}$, where $\beta = \alpha_j$. Its nodes and coefficients are interpreted as their recorded exact dyadic values. The certified trajectory is $\widehat{Z} = I^\alpha \bar{G}$, rather than the floating-point PI nodes themselves. The initial time interval uses the full generalised-power residual expansion. The remaining closed intervals use rigorous convolution-derivative bounds and the mean value theorem, covering all $t \in [0, 1/2]$. At each Fourier node, division of the physical residual by exact ν gives the input to the proved error barrier or certified half-plane linear error estimate.

The recorded exponent is derivative-based: $\bar{L}_T = \nu^{-1} \int \xi_*(T-t)\bar{G}(t)dt$. The positive kernel $q_\alpha = (I^{1-\alpha}\xi_*)' \ge 0$ gives

$$|L_T - \bar{L}_T| \le E_{\mathrm{state}}(I^{1-\alpha}\xi_*)(T)/\nu \le E_{\mathrm{state}}\theta T^{1-\alpha}/[\nu\Gamma(2-\alpha)]. \tag{D.17}$$

No residual integral is added a second time. Gamma and Mittag–Leffler curve moments, trajectory integrals, exponents, phases, and sums for the fixed curve use outward-rounded 100-bit rational intervals.

The positive-half-axis price rule has $h = 1/8$ and analytic-strip width $a = 9/20$. At the available nodes $u = 0 : 1/8 : 64$, the recorded trajectories use their actual residual certificates. For nodes above 64 through 128, the approximate transform is explicitly zero and each error is bounded by the exact-transform envelope. Above 128, the proved infinite discrete-rule tail bound obtained from the proved continuous-integral envelope is included. A comparison bound for the real part of the exact Riccati solution and a 64-cell lower sum over the entire positive-kernel curve give the high-frequency envelope. The analytic-strip trapezoidal error is estimated separately. Passage from the full-axis integral to the positive half-axis, including the half-weight at zero, is handled explicitly. The resulting enclosures cover all twelve model prices in Appendix E and are based on error bounds rather than grid-convergence differences or Richardson diagnostics.

Here E_{state} is a certified state radius: E_δ in Proposition C.5 requires $\delta_F < s_0^2/2$; at other nodes the half-plane bound gives $E_{\mathrm{state}} = \delta_F/\sigma$. The finite-history experiments use Theorem 3.1 and retain the global alternative when valid and tighter. The nearby fields use the same construction with independently generated coefficients and continuous residual banks.

D.3. Nested certificates and the fixed-menu invariant

Proposition D.1 (safe finite refinement). *Fix the output, reference, coefficients, and complete remainder. Replacing a node radius by the smaller of its old radius and a newly proved radius yields a nested joint error set. Consequently every task direction has a nonincreasing certified radius. For a task direction w, choose R_w to bound the remainder support in both directions w and $-w$, and rational U_n to bound the common node coefficient modulus. The symmetric remainder construction used in the numerical examples satisfies this requirement. Let*

$$B_0 \geq R_w + \sum_n U_n \varepsilon_n^0, \quad g = B_0 - R_w - \sum_n U_n \varepsilon_n^0 \geq 0. \tag{D.18}$$

An exact update $B \leftarrow B - U_n(\varepsilon_n^{\text{old}} - \varepsilon_n^{\text{new}})$ preserves the invariant $B = g + R_w + \sum_n U_n \varepsilon_n$. A financial budget is returned as certified only when $DF(|w^\top d| + B) \leq \tau$. An initially sufficient bound needs zero actions. If all finitely many admissible updates suffice and the policy visits every unprocessed action, it terminates in at most that many actions. Otherwise it returns the valid unresolved interval; this does not prove that the actual error exceeds the budget.

Proof. Disk inclusion is preserved by finite products, the pricing map, and addition of the same remainder. Support is monotone under inclusion. The invariant follows by subtracting one nonnegative exact decrement while retaining the initial rounding margin. The stopping and finite-exhaustion statements follow from the invariant and the finite action list. ∎

Choosing the current largest $U_n \varepsilon_n$ is a valid scheduling policy, not a theorem of minimum cost. If all alternative radii have already been certified and every upgrade is assigned unit cost, sorting the exact decrements $U_n(\varepsilon_n^0 - \varepsilon_n^1)$ gives the minimum number of upgrades needed to satisfy a fixed scalar budget. The exchange argument replaces any selected smaller decrement by an unselected larger decrement without reducing progress. The 82-upgrade result in Appendix E concerns only the specified global-maximum-residual finite-history menu; it does not assert optimality for the separately computed time-local radii or other menus. Precomputing the alternative radii is real work and must be included in total cost; minimum action count does not establish minimum resource cost or optimal continuous-residual generation.

D.4. Finite-objective and perturbation proofs

Proof. The exact difference $c_i - M_i$ belongs to the closed interval used in (5.15). The minimum of its square is ℓ and the maximum is v, so the weighted sum encloses every J_j. Each $J_j \geq L_j \geq L_*$; a candidate attaining the smallest U_j gives $J_* \leq J_j \leq U_*$, proving (5.16). If $J_j \leq J_* + \varepsilon$, then $L_j \leq J_j \leq U_* + \varepsilon$, proving (5.17). Conversely, $J_j - J_* \leq U_j - L_*$ gives the stated sufficient inner condition. Under (5.18), every competitor satisfies $J_j - J_{j_0} \geq L_j - U_{j_0} \geq g > 0$, proving uniqueness. Condition (5.19) places the model objective of every selected higher-H candidate above $J_* + \varepsilon$ and therefore excludes it. No independence assumption on quote errors across candidates is needed.

If $p^+ < b^-$, then the exact price satisfies $c < B$; if $p^- > a^+$, then $c > A$. A single such row rules out simultaneous consistency. In the opposite direction, the inner-band conditions give $B \leq b^+ \leq c \leq a^- \leq A$. Applying these statements row by row proves the consistency assertions. ∎

Corollary (objective perturbation). *Suppose another numerical objective $\widetilde{J}$ satisfies $\sup_{\Theta_{\text{finite}}} |\widetilde{J} - J| \leq \delta_J$, and its selected candidate is ε_{alg}-near-optimal. Then*

$$J(\widehat{\alpha}) - J_* \le 2\delta_J + \varepsilon_{\mathrm{alg}}. \tag{D.19}$$

Indeed, $J(\widehat{\alpha}) \le \widetilde{J}(\widehat{\alpha}) + \delta_J \le \min \widetilde{J} + \varepsilon_{\mathrm{alg}} + \delta_J \le J_* + 2\delta_J + \varepsilon_{\mathrm{alg}}$. If the strict gap in (5.18) satisfies $g > 2\delta_J + \varepsilon_{\mathrm{alg}}$, the same unique finite-set minimiser must be selected. This selection stability follows from the finite-objective gap and applies to the fixed candidate sets in Section 7 and Appendix E.

D.5. Legal restriction to another maturity

A continuous field certified on $[0, T_0]$ may be restricted to $[0, T]$, $T \le T_0$, only while retaining the initial value and full history. The derivative-based reference exponent, curve/resolvent weights, reference centre, exact-transform modulus, strip bound and infinite-tail constants must be recomputed for the new maturity. Copying the half-year price centre or simply restarting the fractional equation is invalid. The quarter-year experiment uses this restriction and a separate complete pricing calculation; it does not assert generation of a new quarter-year residual field.

Appendix E. Complete experiments and verification duties

E.1. Three-candidate global-state benchmark

The following price and objective rows use the H0 global-state bank. They have the same fixed quotes and outputs as the three-candidate task. Section 7 compares finite-history propagation and expanded references under the specified configurations; the five-point study uses a separate nearby-candidate bank.

We use twelve quotes at the mathematical maturity $T = 1/2$ from the public SPX sample representing ordinary market conditions. Strikes range from 3700 to 4800 in steps of 100. The original CSV is identified by commit 860049da2b7486fe8aa509061eff23cc28c2ef89 in the WoonJeng data repository. The market bid and ask implied-volatility columns and the authors' model-output columns are identified separately. Market quotes define the objective; the error analysis in this paper supplies the model-price intervals.

The experiment uses normalised call prices with $D = 1, F = 4221.86$. Decimal IV and maturity values in the CSV are treated as exact rational inputs. Standard Black–Scholes conversion of the bid and ask IVs gives B_i, A_i. The target is the price midpoint $M_i = (B_i + A_i)/2$, rather than the Black–Scholes price obtained from the reported midpoint IV. These inputs jointly define the normalised quote setting below.

Fix $\rho = -.7445, \nu = .2897, \lambda_R = \kappa = 0$ and the entire forward variance function

$$\xi_*(t) = .0721 + (.0262 - .0721)E_{.5286}(-.5037t^{.5286}). \tag{E.1}$$

The curve exponent .5286 and curve parameter .5037 remain fixed. Changing α neither recomputes the curve nor assigns .5037 to Riccati mean reversion. These values are the published rounded fitted parameters in [JK2021] and define a candidate comparison with the other inputs fixed.

The candidate set is

$$\Theta_{\mathrm{finite}} = \{13/25, 3/5, 9/10\}, \qquad H = \alpha - \tfrac{1}{2} \in \{.02, .1, .4\}. \tag{E.2}$$

This is a conditional objective profile with all other parameters and the complete curve held fixed. The threshold $H_c = .1$ distinguishes the lower- and higher-H candidates in this comparison. All three satisfy $H < .5$, so the classification is within the roughness

candidates rather than between rough and classical Heston models. All assertions in this section concern the complete three-element set (E.2).

The model prices $c_i(\alpha) = C_i(\alpha)/(DF)$ arise from the admissible nonnegative Volterra variance model and its exact affine transform. The model and kernel assumptions of AbiJaberElEuch2018 are verified individually in this paper; admissibility is established at the probability-model level. Define

$$J(\alpha) = \frac{1}{24}\sum_{i=1}^{12}[c_i(\alpha) - M_i]^2, \qquad J_{\text{band}}(\alpha) = \frac{1}{24}\sum_{i=1}^{12}\operatorname{dist}(c_i(\alpha), [B_i, A_i])^2. \tag{E.3}$$

The normalised price RMS is $\sqrt{2J}$. The same normalisation applies to objective intervals, finite-candidate gaps and stability constants.

Table E.1. H0 half-year three-candidate benchmark: normalized objective, price RMS and actual Padé-output objective enclosures.

α	H	Rigorous model J interval	Rigorous model-price RMS interval	Rigorous implemented Padé J interval
0.52	0.02	[0.0000000068962, 0.0000001833669]	[0.00011744, 0.00060559]	[0.0000000708602, 0.0000000708603]
0.60	0.10	[0.0000004535720, 0.0000006021720]	[0.00095244, 0.00109743]	[0.0000006127366, 0.0000006127367]
0.90	0.40	[0.0000082343900, 0.0000083484458]	[0.00405817, 0.00408619]	[0.0000081393603, 0.0000081393604]

All displayed endpoints are decimal outward roundings of rational bounds; the decisions use the untruncated exact rational endpoints. The unique minimiser of the model objective is α=0.52, H=0.02. A strict lower bound on the gap between its objective and those of its competitors is 0.0000002702052. The implemented Padé output has the same unique finite-set minimiser, as certified by strict interval separation.

For ε=0, the outer set of exact optimal candidates is $\{13/25\}$. The specified higher-H candidates with α≥.6 are strictly separated from this optimum.

Table E.2. Historical three-candidate quote-band objective and consistency decisions, using the original twelve quotes.

α	Rigorous model J_band interval	Rows strictly excluding quote consistency	All-row quote consistency certified
0.52	[0.0000000005732, 0.0000000877280]	[8, 9]	No
0.60	[0.0000003444775, 0.0000004648236]	[5, 6, 7, 8, 9, 10, 11]	No
0.90	[0.0000076345237, 0.0000077422463]	[1, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12]	No

Least-squares ordering and quote consistency are determined separately from objective intervals and rowwise price intervals. The following tables report the prices supporting these two model-validation results.

α=0.52, H=0.02

Table E.3. H0 alpha=0.52 price accounts: normalized quotes, model prices, actual output and its total error bound.

K	Rigorous quote-midpoint interval	Rigorous normalised model-price interval	Implemented Padé dyadic output (displayed value)	Total model-price error bound for this output
3700	[0.13916863, 0.13916864]	[0.13882872, 0.13937153]	0.13910887	0.00028016
3800	[0.11921230, 0.11921231]	[0.11858902, 0.11913912]	0.11888921	0.00030019
3900	[0.09981335, 0.09981336]	[0.09909480, 0.09965209]	0.09943278	0.00033799
4000	[0.08123198, 0.08123199]	[0.08055552, 0.08111992]	0.08092667	0.00037115
4100	[0.06371452, 0.06371453]	[0.06322696, 0.06379836]	0.06361496	0.00038801
4200	[0.04759685, 0.04759686]	[0.04740368, 0.04798201]	0.04781654	0.00041287
4300	[0.03333802, 0.03333803]	[0.03349166, 0.03407684]	0.03393723	0.00044558
4400	[0.02173166, 0.02173167]	[0.02200137, 0.02259332]	0.02244498	0.00044361
4500	[0.01316937, 0.01316938]	[0.01332823, 0.01392686]	0.01373568	0.00040746
4600	[0.00760335, 0.00760336]	[0.00746696, 0.00807221]	0.00785225	0.00038529
4700	[0.00434644, 0.00434645]	[0.00393703, 0.00454881]	0.00430503	0.00036801
4800	[0.00253419, 0.00253420]	[0.00199815, 0.00261641]	0.00232396	0.00032582

α=0.60, H=0.10

Table E.4. H0 alpha=0.60 price accounts, with the same normalization and column meanings as Table E.3.

K	Rigorous quote-midpoint interval	Rigorous normalised model-price interval	Implemented Padé dyadic output (displayed value)	Total model-price error bound for this output
3700	[0.13916863, 0.13916864]	[0.13908096, 0.13925890]	0.13914793	0.00011096
3800	[0.11921230, 0.11921231]	[0.11900101, 0.11918134]	0.11908020	0.00010114
3900	[0.09981335, 0.09981336]	[0.09970355, 0.09988623]	0.09981101	0.00010746
4000	[0.08123198, 0.08123199]	[0.08138513, 0.08157015]	0.08152385	0.00013873
4100	[0.06371452, 0.06371453]	[0.06429081, 0.06447812]	0.06445106	0.00016025
4200	[0.04759685, 0.04759686]	[0.04869874, 0.04888832]	0.04888577	0.00018704

K	Rigorous quote-midpoint interval	Rigorous normalised model-price interval	Implemented Padé dyadic output (displayed value)	Total model-price error bound for this output
4300	[0.03333802, 0.03333803]	[0.03496352, 0.03515534]	0.03518748	0.00022397
4400	[0.02173166, 0.02173167]	[0.02351737, 0.02371142]	0.02375739	0.00024002
4500	[0.01316937, 0.01316938]	[0.01471126, 0.01490749]	0.01493438	0.00022313
4600	[0.00760335, 0.00760336]	[0.00857862, 0.00877702]	0.00878309	0.00020447
4700	[0.00434644, 0.00434645]	[0.00474038, 0.00494092]	0.00492883	0.00018845
4800	[0.00253419, 0.00253420]	[0.00255071, 0.00275338]	0.00270300	0.00015230

α=0.90, H=0.40

Table E.5. H0 alpha=0.90 price accounts, with the same normalization and column meanings as Table E.3.

K	Rigorous quote-midpoint interval	Rigorous normalised model-price interval	Implemented Padé dyadic output (displayed value)	Total model-price error bound for this output
3700	[0.13916863, 0.13916864]	[0.13880016, 0.13883098]	0.13876501	0.00006596
3800	[0.11921230, 0.11921231]	[0.11929811, 0.11932934]	0.11923074	0.00009859
3900	[0.09981335, 0.09981336]	[0.10073709, 0.10076873]	0.10063802	0.00013071
4000	[0.08123198, 0.08123199]	[0.08327826, 0.08331031]	0.08315433	0.00015597
4100	[0.06371452, 0.06371453]	[0.06710772, 0.06714016]	0.06696694	0.00017321
4200	[0.04759685, 0.04759686]	[0.05242401, 0.05245684]	0.05228161	0.00017523
4300	[0.03333802, 0.03333803]	[0.03943228, 0.03946550]	0.03931664	0.00014886
4400	[0.02173166, 0.02173167]	[0.02834450, 0.02837811]	0.02828552	0.00009258
4500	[0.01316937, 0.01316938]	[0.01934300, 0.01937699]	0.01935637	0.00002061
4600	[0.00760335, 0.00760336]	[0.01249183, 0.01252619]	0.01257890	0.00008707
4700	[0.00434644, 0.00434645]	[0.00765389, 0.00768862]	0.00780424	0.00015035
4800	[0.00253419, 0.00253420]	[0.00449079, 0.00452590]	0.00467714	0.00018635

The Padé procedure uses the six-condition GR [3/3] approximation, a 256-point Jacobi rule for the derivative-based exponent, and an eighth-order composite Gauss–Legendre rule for Fourier inversion through 200. The archived files identify the Python/NumPy implementation and source hashes. Each finite output c^P is saved as an exact dyadic rational. The model-price reference interval gives

$$|c^P - c| \leq \max\{|c^P - c^-|, |c^P - c^+|\}. \tag{E.4}$$

This bound includes the implementation's state, exponent, quadrature, truncation, and floating-point errors. The saved outputs can therefore be compared directly without a separate certification of an infinite-integral analytic Padé price. The numerical objective uses the same quote-price midpoint intervals. We assert selection preservation only when both the numerical and model objectives are strictly separated on the finite set and have the same minimiser. The assertion concerns the specified implementation, candidates, and input setting.

The H0 benchmark uses the uniform-state barrier in Appendix C.1. Section 3 gives finite-history and resolvent propagation; Section 7 compares them on the identified configurations.

E.2. Numerical identities, scope and evidence map

In this matrix, IDs label proof obligations. They are separate from the configuration IDs in Section 7; obligation N2 covers the N2L local-bin refinement.

`configuration-registry.json` fixes the configuration IDs and mathematical dimensions. The matrix lists what each reader recomputes and which interval primitives, generator bounds and saved inputs it shares. The acceptance records document the author's runs.

Table E.6. Proof obligations, reader locations, recomputation and shared dependencies.

ID	Obligation	Verification command	Recomputed	Inherited / shared
S1	Padé coefficient signs on the original parameter rectangle	`full_structure_verify.py` (S1)	211241 structural sign leaves and rectangle cover	model/ parameter definitions; original strict interval algebra and generalized-power primitives
S2	Startup Caputo residual and left-halfplane enclosure	`startup_independent.py` (S2)	all 513 frequencies for each of the three original startup cells	stored field dyadics and later-time continuous residuals; Gamma/ power/ dyadic outward primitives
R1	Full original alpha=.52 low-frequency continuous residual cover	`check-time-envelope.py` (R1)	8189 closed cells ×513 entries: cover, saved dyadic values and per-node maxima; startup audited separately	derivative inequalities in identified continuous generator; NumPy enclosure with proved dot-product error and strict scalar primitives

ID	Obligation	Verification command	Recomputed	Inherited / shared
R2	Full alpha=.52 high-frequency continuous residual cover	`omission-verify.py` (R2)	8189 closed cells ×512 entries: exact bank, startup/ halfplane, saved-node maxima	identified Caputo derivative-generat or inequalities; strict scalar primitives and original continuous generator
P1	Global finite-history and low-frequency 128-bin price accounts	`verify-finite-history-independent.py`; `independent-time-local.py` (P1)	forward masses, complete cell intersections, 513 tightened radii, all 1025 finite terms and complete spread remainder	continuous residual theorem/ validity and identified stored field; strict forward-moment, exponent and true-tail primitives
P2	Expanded-reference frozen half-year output account	`independent-omission.py` (P2)	changed reference centre, all 1025 support terms, old/ new complete interval intersection, global/ local high-frequency accounts	same actual fast output and continuous residual banks; original strict moment/ scalar primitives
Q1	Quarter-maturity used 64 certificate and actual fast output	`independent-transfer.py` (Q1)	all three exact terminal interpolations, 1539 restricted exponents, three original fast-output calculations, all 3075 true finite CF/ tail/ price terms	larger-horizon continuous residuals and halfplane certificate; strict moment/ trig/ dyadic/ true-tail libraries; original fast implementation used only for output replay
Q2	Quarter full 128 global and 128-bin complete certificate	`independent-transfer-supplement.py` (Q2)	all 1025 restricted exponents, all 1025 CF/ tail/ coefficient/ radius terms, all high bank entries, all 128 local weights and exact account	low/ high continuous derivative validity; actual fast replay is separately performed by Q1; strict moment/ trig/ dyadic/ true-tail libraries; path helpers of Q1
N1	Fresh fields/ residuals at the five nearby alpha points	`nearby_independent.py` (N1)	all cells, startup, halfplane, 513 exponents and 1025 CF/ coefficient terms per point; all objective/ pair decisions	strict proof of continuous residual generator and final propagation theorem; declared sdk outward primitives, stored reference data

ID	Obligation	Verification command	Recomputed	Inherited / shared
N2	Descriptive 128-bin reused-bank control, outside nearby objective grid	`local_output_independent.py` (N2)	128 exact positive weights, every intersecting closed-bank maximum, 513 tightened radii, all 12 prices and all actual output rounding	complete identity-matched N2048 point proof and continuous residual generator; same strict moment/ trig/ dyadic primitives
C1	Same model and complete tolerance for frozen/ direct/ corrected/ BL-core outputs	`workload_controls_independent.py` (C1)	exact stored-output centre translation, all same-reference remainder and 28 portfolio supports; BL nominal-output metadata identity	reference certificate, BL nominal-output producer and its stored dyadics; same strict outward coefficient primitives

The command index gives copyable commands and working directories. `proof-obligation-matrix.json` records generation behavior and source hashes. Run regeneration in a disposable work copy. `nearby_generate.py` generates missing fields and residuals and reuses matching existing components. `nearby_independent.py` permits an identity-matched per-point cache; the complete acceptance driver deletes those caches before calling it.

The complete entry point and continuous-regeneration commands have the following scopes:

Table E.7. Bank reading, complete structural checks and continuous-regeneration commands.

Entry point	Additional executed obligation	Outside that command
`reproduce.py`	saved-bank readers and downstream reconstruction; both quarter readers run with --full and nearby point caches are cleared	continuous residual generation and the complete structural sign cover
`reproduce.py --full`	the same readers plus all 211241 structural sign leaves and the supplementary checks in Table E.10	fresh low/ high/ nearby continuous derivative generation
`run_evidence.py --regenerate-continuous`	alpha=.52 513-node low continuous generator and 8189-cell bank	five-alpha reference/ residual generation
`run_frontier.py --regenerate-full`	separate-copy 512-node high continuous generator and fresh-bank audit	changing the fixed quarter inputs
`nearby_generate.py --N N`	missing field/ residual/ exponent components; matching existing components are reused	forced regeneration over existing scientific output
`verify_source.py`	published source hashes	bank replay, interval reconstruction and residual generation

`reproduce.py --full` checks the complete structural cover as well as saved-bank reconstruction. Regenerating continuous derivative bounds requires the dedicated commands. The low-frequency route uses `run_evidence.py --regenerate-continuous`; the high-frequency route uses `run_frontier.py --regenerate-full` in a separate reconstruction

copy. This preserves the fixed quarter inputs. The source check establishes file identity, while the readers establish the recomputed obligations in the matrices.

The exact closest-pair gap equals the sum of the following signed contributions; scientific notation is a display only:

Table E.8. Signed contributions to the closest-pair normalized objective gap; scientific notation displays the exact ledger.

Gap contribution	Normalized loss contribution
reference_midpoint_objective_difference	7.939186982887e-09
used_node_support_both_candidates	-4.690368762320e-09
finite_zero_support_both_candidates	-1.217354923855e-09
strip_both_candidates	-1.855578458131e-12
true_infinite_tail_both_candidates	-9.568024410400e-11
reference_arithmetic_both_candidates	-6.928606526135e-23
midpoint_conversion_arithmetic_both_candidates	-4.770477111773e-27
candidate_a_quadratic_remainder	-1.195663126282e-09
box_intersection_endpoint_adjustment	0.000000000000e+00

The quarterly account has zero finite omissions; its infinite tail beyond 128 remains paid. The unused-node term in the nearby/global controls is nonzero and cannot be dropped by borrowing the quarter full-reference result. Exact ledger fractions and all 40 pair-mode records are delivered with independent secondary readback and intentional corruption controls.

Common ideal strict-reference floor in the BL-core complete bound (floors and percentages are approximate displays; exact fractions are authoritative):

Table E.9. Common strict-reference floor and output translation in BL-core certificates; price quantities are in index points.

Bank	Nominal output	Reference floor points	Translation points	Floor / complete bound
N1	BL modified-Adams core, 512 steps	0.423484833	0.000853401	99.798887%
N1	BL modified-Adams core, 1024 steps	0.423484833	0.000305328	99.927953%
N2	BL modified-Adams core, 512 steps	0.261808768	0.000845646	99.678039%
N2	BL modified-Adams core, 1024 steps	0.261808768	0.000297572	99.886469%
N2L	BL modified-Adams core, 512 steps	0.228237113	0.000845646	99.630856%
N2L	BL modified-Adams core, 1024 steps	0.228237113	0.000297572	99.869791%

The separate returned-reference ratio also pays binary64 return rounding and is retained in the machine ledger. A small BL512/1024 nominal difference is an empirical comparison; its complete guarantee uses the same reference floor plus exact output translation. These floor fractions do not establish an intrinsic BL error estimate.

`portfolio-pass-counts.svg` and `portfolio-pass-counts.pdf` plot exact-endpoint pass counts for the three candidates and the alpha=.52 nearby controls. Coordinates are converted to decimals after exact counting. The shown range is 0–2 index points; the CSV/JSON retains

every threshold. Within-bank comparisons use matched radii, while cross-bank curves describe different configurations.

The complete driver also runs the dissipative-kernel reader and exact experimental-ledger reader. These complement the bank reconstruction and reuse the upstream continuous derivative bounds.

Table E.10. Supplementary readers: recomputed obligations and shared inputs.

Obligation	Generate	Verify	Recomputed / shared inputs
Dissipative weights and complete prices	`generate.py` (K1)	`independent.py` (K1)	14-layer direct series versus 12-layer moments; all 128 bins, 1025 nodes, 12 prices and 8 controls; strict primitives and upstream residual proof shared
Exact experiment ledgers and thresholds	`produce_audit.py` (A1)	`read_audit_independent.py` (A1)	Exact saved-bank accounts, pair gaps, 504 thresholds and 9 controls; inherited derivative validity

E.3. Budget sensitivity of every fixed direction

Figure E.1 uses exact endpoints for all 28 declared half-year directions. Solid curves denote joint certificates and dashed curves signed-marginal certificates; radii are matched within each configuration. Comparisons between different banks are descriptive. The shown range is zero to two index points; the machine table retains every threshold, including those beyond this plotting range.

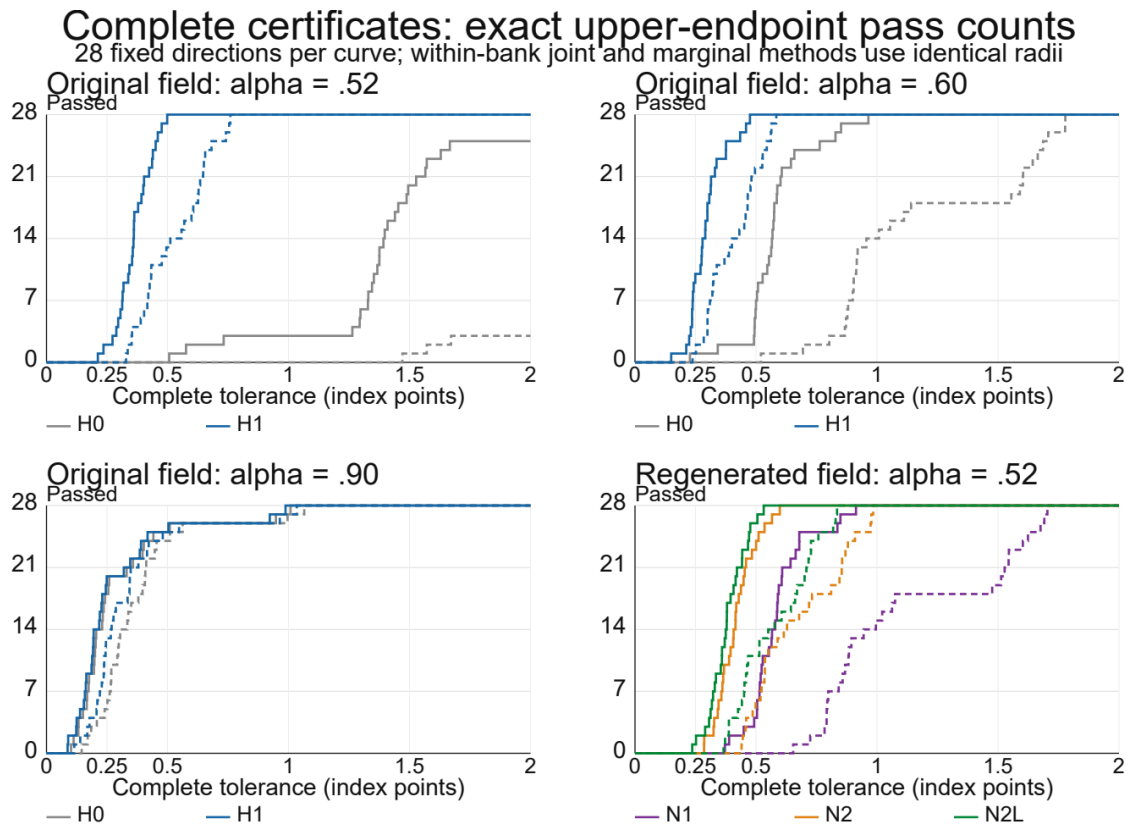


Figure E.1. Exact-threshold budget–pass-count curves; 28 fixed directions per configuration.

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